

Continuous Gaussian Hidden Markov Models for Equity and Volatility Index Dynamics: A Baum-Welch Approach Without Jump Augmentation

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Abstract

We present a continuous hidden Markov model (CHMM) with Gaussian emissions trained via the Baum-Welch (Expectation-Maximization) algorithm for generating synthetic financial time series that preserve the canonical stylized facts of asset returns. Unlike the discrete-state framework of Alswaidan and Varner [2026], which requires Laplace quantile binning and a Poisson jump-duration mechanism to achieve volatility clustering, we show that the continuous model at moderate state resolution ($K = 6\text{--}13$) reproduces all three stylized facts—heavy-tailed distributions, negligible linear autocorrelation, and persistent volatility clustering—without jump augmentation, thereby eliminating two tunable hyperparameters (ϵ, λ) and the discretization step entirely.

We benchmark the CHMM against five alternative generators on ten years of daily S&P 500 ETF (SPY) data: bootstrap resampling, Gaussian and Laplace i.i.d. generators, GARCH(1,1), and the discrete HMM with jumps from Alswaidan and Varner [2026]. Evaluation uses seven complementary metrics—Kolmogorov–Smirnov and Anderson–Darling pass rates, excess kurtosis, autocorrelation function mean absolute error (ACF-MAE), Wasserstein-1 and Hellinger distances, and quantile coverage—applied to 1,000 simulated paths in both in-sample (2014–2024) and out-of-sample (2025) windows. Cross-asset generalization to NVDA, JNJ, and JPM confirms adaptability to diverse risk profiles.

We further extend the framework to CBOE Volatility Index (VIX) and VXX log returns, demonstrating that the CHMM captures the distinct statistical properties of volatility indices—including stronger mean reversion, positive skewness, and pronounced regime structure. The VIX regime decomposition provides a natural bridge to option pricing: median VIX levels per decoded regime define a volatility map that drives regime-switching geometric Brownian motion paths, producing implied volatility surfaces that we compare against the Heston stochastic volatility benchmark.

Keywords: Continuous Hidden Markov Model, Baum-Welch Algorithm, Stylized Facts, Volatility Clustering, Regime-Switching, Synthetic Financial Data, VIX, Option Pricing

1 Introduction

Generating synthetic financial time series that faithfully preserve the statistical properties of real market data is a central challenge in quantitative finance. High-fidelity synthetic data are essential for stress testing risk models against plausible but unobserved market scenarios, validating portfolio optimization algorithms, and augmenting limited training sets for machine learning pipelines [Assefa et al., 2020, Jordon et al., 2022]. Empirical equity excess growth rates

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exhibit three well-documented statistical regularities—heavy-tailed (leptokurtic) distributions, negligible linear autocorrelation in raw returns, and persistent volatility clustering—that any generative framework must simultaneously reproduce to be considered statistically faithful [Cont, 2001, Mandelbrot, 1963].

Parametric approaches to synthetic data generation face a fundamental tension between distributional accuracy and temporal fidelity. GARCH models [Engle, 1982, Bollerslev, 1986] excel at reproducing the slow decay of the autocorrelation function (ACF) of absolute returns—the hallmark of volatility clustering—but their unimodal innovation distribution cannot represent the multi-regime structure of equity markets, leading to poor distributional fit. Conversely, i.i.d. generators that match the marginal distribution (e.g., Laplace sampling) achieve high distributional pass rates but produce flat ACF profiles by construction. Hidden Markov models (HMMs) offer a middle path: the latent regime structure can simultaneously generate heavy-tailed mixtures and temporal persistence through the transition dynamics.

However, the application of HMMs to financial stylized facts has been constrained by an influential negative result. Rydén et al. [1998] systematically evaluated Gaussian-emission HMMs with $K = 2\text{--}3$ states against the canonical stylized facts and concluded that while HMMs reproduce heavy tails and negligible return autocorrelation, they *fail* to capture the slow decay of the ACF of absolute returns. The mechanism is clear: with few states, the geometric sojourn-time distribution of a standard Markov chain produces exponentially fast ACF decay, insufficient to match the hyperbolic-like persistence observed empirically. Bulla and Bulla [2006] subsequently showed that hidden semi-Markov models (HSMMs), which replace the geometric sojourn distribution with flexible alternatives, restore ACF fidelity—but at the cost of substantially increased model complexity.

Alswaidan and Varner [2026] proposed an alternative solution within the discrete HMM framework: a Poisson jump-duration mechanism that forces the model to dwell in tail states for empirically calibrated durations. That approach achieved 91–95% Kolmogorov–Smirnov (KS) pass rates across 424 US equities and partially reproduced volatility clustering. However, the discrete framework introduced three sources of complexity: (i) Laplace quantile binning, which discretizes continuous observations and introduces quantization error; (ii) two additional hyperparameters (ϵ for jump probability and λ for mean jump duration) requiring grid search calibration; and (iii) within-bin Student- t emission models whose parameters are estimated separately from the transition structure.

In this paper, we show that these complexities are unnecessary. A continuous Gaussian HMM (CHMM) trained via the Baum-Welch algorithm [Baum et al., 1970, Rabiner and Juang, 1986] with quantile-based initialization reproduces all three canonical stylized facts at moderate state resolution ($K = 6\text{--}13$) *without* requiring jump augmentation, discretization, or separate emission fitting. The key insight is that the negative result of Rydén et al. [1998] is an artifact of insufficient state resolution: at $K \geq 3$ with proper initialization, the Baum-Welch algorithm learns a transition matrix with sufficient diagonal dominance in high-volatility states to generate slow ACF decay, effectively approximating the flexible sojourn-time distributions of HSMMs through a mixture of geometric distributions.

The contributions of this work are:

1. **Baum-Welch training without jump augmentation.** We train continuous Gaussian emission parameters (μ_k, σ_k) and the transition matrix $\mathbf{T}$ jointly from raw observations via the Baum-Welch algorithm with quantile-based initialization. The resulting CHMM achieves 90–94% KS pass rates and ACF-MAE of 0.046–0.049 on absolute returns—comparable to GARCH(1,1) (ACF-MAE = 0.047)—without any jump mechanism, eliminating two hyperparameters (ϵ, λ) and the discretization step.
2. **Contradiction of the Rydén et al. (1998) consensus.** We demonstrate that the widely cited limitation of Gaussian HMMs—inability to reproduce volatility clustering—is

a consequence of testing only $K = 2\text{--}3$ states. At moderate K , the learned transition matrix develops sufficient persistence in high-volatility states to achieve ACF-MAE comparable to GARCH, which we show is effectively an approximation to the HSMM solution of Bulla and Bulla [2006].

3. **Comprehensive five-model benchmark.** We evaluate the CHMM against bootstrap resampling, Gaussian and Laplace i.i.d. generators, and GARCH(1,1) using seven complementary metrics—KS and Anderson–Darling (AD) pass rates, excess kurtosis, ACF-MAE, Wasserstein-1 and Hellinger distances, and quantile coverage—on 1,000 simulated paths in both in-sample (2014–2024) and out-of-sample (2025) windows.
4. **State resolution analysis.** We systematically vary $K \in \{3, 6, 9, 11, 13\}$ and show that distributional fidelity improves with K while ACF-MAE remains remarkably stable, and quantile coverage is 100% at all resolutions.
5. **Cross-asset and volatility index generalization.** We confirm robustness across NVDA, JNJ, and JPM (97–100% IS KS), and extend the framework to CBOE VIX and VXX log returns, demonstrating that the CHMM captures the distinct statistical properties of volatility indices and provides a natural bridge to regime-switching option pricing.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on stylized facts, parametric models, HMMs, and deep generative approaches. Section 3 describes the data, the CHMM specification, the Baum–Welch training procedure, the benchmark models, and the validation metrics. Section 4 presents the empirical results: descriptive statistics, five-model comparison, state resolution sensitivity, out-of-sample evaluation, cross-asset generalization, and the VIX/VXX extension. Section 5 interprets the findings—particularly the ACF-MAE result and its implications for the Rydén et al. consensus—and identifies limitations. Section 6 summarizes the contributions and outlines future directions.

2 Related Work

Research on reproducing the canonical stylized facts of financial markets spans parametric return models, latent-state frameworks, and deep generative approaches. This section reviews the key developments that motivate the present work, with particular attention to the Gaussian HMM limitation identified by Rydén et al. [1998] that our results challenge.

2.1 Stylized Facts of Asset Returns

The Black–Scholes–Merton framework [Black and Scholes, 1973] assumes that log-returns are independently and identically distributed (i.i.d.) Gaussian random variables. This assumption, while analytically convenient, is contradicted by a large body of empirical evidence accumulated over six decades. Mandelbrot [1963] demonstrated that speculative price changes exhibit heavy tails (leptokurtosis), with extreme returns occurring far more frequently than a Gaussian model predicts. Fama [1970] confirmed negligible linear autocorrelation in raw returns, consistent with informational efficiency, but this does not imply independence: nonlinear dependence persists in the form of volatility clustering. Schwert [1989] documented that the autocorrelation of absolute returns decays slowly, often following a hyperbolic rather than exponential pattern, providing the empirical signature of clustered volatility.

Cont [2001] systematized these observations into a canonical set of stylized facts that hold across asset classes, geographic markets, and sampling frequencies:

1. *Heavy tails:* the marginal distribution of returns exhibits positive excess kurtosis, with power-law tail decay.

2. *Absence of linear autocorrelation*: raw returns are approximately uncorrelated, consistent with market efficiency.
3. *Volatility clustering*: the autocorrelation of absolute (or squared) returns is positive and decays slowly, indicating that large price movements tend to be followed by large movements of either sign.

Any generative model for financial time series must simultaneously reproduce all three facts to be considered statistically faithful. Failure to capture even one—as we will show for the Gaussian i.i.d. generator (distributional failure) and GARCH(1,1) (distributional failure despite temporal fidelity)—renders the synthetic data inadequate for downstream applications such as risk measurement and portfolio backtesting.

2.2 Parametric Volatility Models

The ARCH model of Engle [1982] introduced conditional heteroscedasticity into time series econometrics, modeling the variance of returns as a function of past squared residuals. The generalized ARCH (GARCH) extension of Bollerslev [1986] added lagged conditional variance, producing the workhorse GARCH(1,1) specification:

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (1)$$

where $\alpha + \beta < 1$ ensures stationarity. The single-decay parameter β directly controls the persistence of the conditional variance process, producing exponential ACF decay that closely approximates the empirical pattern when $\alpha + \beta$ is near unity. However, GARCH models assume a unimodal innovation distribution and cannot represent discrete market regimes. As we demonstrate empirically, this unimodal assumption leads to catastrophic distributional failure: GARCH(1,1) achieves only 13% KS pass rate despite matching the observed ACF-MAE almost exactly (0.047 vs. 0.048 observed).

Stochastic volatility (SV) models treat the log-variance as a latent process. The Heston [1993] model specifies the variance as a mean-reverting square-root diffusion:

$$dv_t = \kappa(\theta - v_t) dt + \sigma_v \sqrt{v_t} dW_t^v \quad (2)$$

correlated with the price diffusion via $\text{Corr}(dW_t^S, dW_t^v) = \rho$. SV models produce realistic volatility dynamics and closed-form option prices, but parameter estimation requires computationally intensive methods (e.g., particle filtering, MCMC). The Heston model features prominently in our VIX extension, where we use CHMM-decoded regimes to parameterize the mean-reversion target θ .

Jump-diffusion models [Merton, 1976, Kou, 2002] augment the geometric Brownian motion with Poisson-driven jumps in the price level, producing heavy-tailed marginal distributions. Merton [1976] used Gaussian jump sizes, while Kou [2002] introduced double-exponential jumps to capture the asymmetry between upside and downside tail risk. These models generate heavy tails but lack a mechanism for volatility persistence: after a jump event, the volatility returns immediately to its baseline level, failing to reproduce the clustering phenomenon.

2.3 Hidden Markov Models and the Rydén et al. Limitation

Hamilton [1989] introduced the Markov-switching model to economics, demonstrating that U.S. GNP growth is well described by a two-state regime-switching process. The model posits a discrete latent state S_t that evolves as a first-order Markov chain, with the observable variable drawn from a state-dependent distribution. HMMs have since been applied to financial time series for regime identification [Kim and Nelson, 1999], volatility estimation [Rossi and Gallo, 2006], and trading strategy design [Nguyen, 2018].

The seminal evaluation of HMMs against stylized facts is due to [Rydén et al. \[1998\]](#), who fitted Gaussian-emission HMMs with $K = 2$ and $K = 3$ states to daily Swedish equity returns. They established three key findings:

1. HMMs produce heavy-tailed marginal distributions through the Gaussian mixture mechanism, even though each component is thin-tailed.
2. HMMs produce negligible linear autocorrelation in returns, consistent with the observed stylized fact.
3. HMMs *fail* to reproduce the slow decay of the ACF of absolute returns (volatility clustering).

The third finding follows from a structural property of Markov chains: the sojourn time in any state follows a geometric distribution, which implies exponential decay of the ACF. For $K = 2-3$ states, the exponential decay is too fast to approximate the hyperbolic-like persistence observed empirically. This result has been widely cited as a fundamental limitation of standard HMMs for financial modeling.

[Bulla and Bulla \[2006\]](#) proposed hidden semi-Markov models (HSMMs) as a remedy. HSMMs replace the geometric sojourn-time distribution with flexible parametric alternatives (e.g., negative binomial, log-series), allowing arbitrarily slow ACF decay. They demonstrated that HSMMs with $K = 2-4$ states and negative-binomial sojourn distributions successfully reproduce all three stylized facts, confirming that the Rydén limitation is fundamentally about the sojourn-time distribution, not the HMM framework itself.

The present work offers an alternative resolution: rather than modifying the sojourn-time distribution, we increase the state resolution. With $K \geq 3$ states and quantile-based initialization, the Baum-Welch algorithm learns a transition matrix where high-volatility states exhibit sufficient self-transition probability (diagonal dominance) to sustain persistence. The resulting mixture of K geometric sojourn-time distributions—each with a different rate parameter—can approximate slow ACF decay, effectively achieving an HSMM-like effect within the standard HMM framework. This insight has not been explored in the prior literature, likely because [Rydén et al. \[1998\]](#) only tested $K = 2-3$ and subsequent work adopted the HSMM path.

2.4 Discrete HMM with Jump Augmentation

[Alswaidan and Varner \[2026\]](#) addressed the Rydén limitation through a different mechanism: a Poisson jump-duration process that forces the model to dwell in tail states for durations drawn from a Poisson distribution, overriding the geometric sojourn time. Their discrete framework used Laplace quantile binning to partition continuous observations into categorical states, estimated the transition matrix by direct frequency counting, and employed Student- t within-bin emission distributions. The jump mechanism introduced two hyperparameters (ϵ for jump probability, λ for mean duration) calibrated via grid search. The framework achieved 91–95% KS pass rates across 424 US equities and partially reproduced volatility clustering.

The present work eliminates all three sources of complexity introduced by the discrete framework: (i) continuous Gaussian emissions replace Laplace quantile binning, removing discretization artifacts; (ii) the Baum-Welch algorithm jointly learns emissions and transitions, replacing frequency counting; (iii) no jump mechanism is needed at moderate K , removing two hyperparameters and the grid search entirely.

2.5 Deep Generative Models

Deep generative models have attracted growing interest for financial time series synthesis. [Takahashi et al. \[2019\]](#) applied generative adversarial networks (GANs) to stock return generation,

finding that GAN-produced sequences visually resembled financial data but failed to reproduce ACF structure—the same volatility-clustering limitation as low- K HMMs. [Kwon and Lee \[2024\]](#) systematically evaluated several GAN architectures against the Cont stylized facts and confirmed that volatility clustering remained the most difficult property to capture, unless the architecture was specifically designed to encode temporal dependence (e.g., through recurrent discriminators).

[Alswaidan and Varner \[2026\]](#) benchmarked a GRU-based generator against HMM variants and found that it closed 83% of the ACF-MAE gap relative to the HMM, but suffered variance collapse (0.6% KS pass rate), indicating that distributional fidelity and temporal structure remain in tension for neural approaches. The present study focuses on the parametric HMM approach, which offers interpretability (each state corresponds to a named market regime), computational efficiency ($O(TK^2)$ per Baum-Welch iteration), and guaranteed convergence (EM monotonicity), advantages that complement the flexibility of deep generative methods.

2.6 Quality Assessment for Synthetic Time Series

[Stenger et al. \[2024\]](#) surveyed the quality assessment landscape for synthetic time series, organizing evaluation metrics into distributional fidelity, temporal fidelity, and downstream utility categories. Our seven-metric evaluation framework—KS and AD pass rates, excess kurtosis, ACF-MAE, Wasserstein-1, Hellinger distance, and quantile coverage—spans the first two categories, providing a comprehensive assessment that avoids reliance on any single metric. As we will demonstrate, no single model dominates all seven metrics, and the choice of generator depends on which dimension of fidelity is prioritized. [Assefa et al. \[2020\]](#) made a complementary argument in the specific context of financial data, emphasizing that synthetic generators should be evaluated on their ability to preserve statistical properties relevant to the intended downstream application.

3 Methods

3.1 Data

The empirical foundation of this study is a dataset of 424 United States-listed equities and exchange-traded funds spanning ten years of daily trading data (January 3, 2014 to December 31, 2024). We focus the primary single-asset analysis on the SPDR S&P 500 ETF Trust (SPY), which tracks the S&P 500 index and is the most liquid equity ETF by trading volume. The data include daily open, high, low, and close prices along with volume and volume-weighted average price (VWAP) metrics, sourced from standard financial data providers. The in-sample (IS) dataset comprises $T_{\text{IS}} = 2,766$ trading days. A subsequent $T_{\text{OoS}} = 219$ trading days in 2025 are reserved as a held-out out-of-sample (OoS) test set to evaluate generalization.

Cross-asset generalization is assessed on three additional equities spanning distinct risk profiles: NVDA (NVIDIA Corporation; high-beta technology), JNJ (Johnson & Johnson; low-beta healthcare), and JPM (JPMorgan Chase & Co.; moderate-beta financials). Each asset is fitted independently using the same training/test split.

To extend the framework beyond equity returns, we additionally apply it to the CBOE Volatility Index (VIX), which measures the 30-day implied volatility of S&P 500 options and is widely regarded as the market’s “fear gauge.” VIX daily close prices spanning approximately 2006–2024 constitute the in-sample dataset, with 2025 data reserved for out-of-sample testing. VIX log returns exhibit statistical properties that differ markedly from equity returns—positive skewness (reflecting asymmetric volatility spikes), pronounced mean reversion, and extreme excess kurtosis—making it a challenging test case that probes the generality of the CHMM framework.

3.2 Excess Growth Rate Calculation

We compute the annualized excess growth rate $G_{i,j}$ for ticker i between trading days $j - 1$ and j given the equity price series $P_{i,j}$:

$$G_{i,j} \equiv \frac{1}{\Delta t} \cdot \ln\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f \quad (3)$$

where $\Delta t = 1/252$ (one trading day expressed in years) and r_f is a constant risk-free rate set to zero throughout this study. The resulting excess growth rate has units of year^{-1} and is time-additive under continuous compounding. The annualization by $1/\Delta t$ scales the daily log-returns to an annualized basis, producing values with magnitudes on the order of ± 2 standard deviations for typical daily moves.

3.3 Continuous Hidden Markov Model

We define the continuous hidden Markov model (CHMM) as the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{B}, \boldsymbol{\pi})$ [Rabiner and Juang, 1986], where:

- $S_t \in \mathcal{S} = \{1, 2, \dots, K\}$ is the hidden state at time t , representing the prevailing market regime;
- $\mathcal{O} \subseteq \mathbb{R}$ is the continuous observation space of excess growth rates;
- $\mathbf{T} \in \mathbb{R}^{K \times K}$ is the state transition probability matrix with entries $T_{ij} = \mathbb{P}(S_{t+1} = j \mid S_t = i)$, satisfying $\sum_{j=1}^K T_{ij} = 1$ for all i ;
- $\mathbf{B} = \{(\mu_k, \sigma_k)\}_{k=1}^K$ specifies the Gaussian emission distributions, where $G_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k^2)$;
- $\boldsymbol{\pi} = (\pi_1, \dots, \pi_K)$ is the initial state distribution with $\pi_k = \mathbb{P}(S_1 = k)$.

The emission density for state k is:

$$b_k(x) = \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma_k^2}\right) \quad (4)$$

The marginal distribution of G_t under the stationary regime is a K -component Gaussian mixture:

$$f(x) = \sum_{k=1}^K \bar{\pi}_k b_k(x) \quad (5)$$

where $\bar{\boldsymbol{\pi}}$ is the stationary distribution satisfying $\bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}} \mathbf{T}$. This mixture structure is the mechanism by which HMMs produce heavy-tailed marginal distributions from thin-tailed Gaussian components: tail states with large $|\mu_k|$ or σ_k contribute to the kurtosis of the mixture even though each component has zero excess kurtosis individually.

Unlike the discrete framework of Alswaidan and Varner [2026], which discretizes observations into Laplace quantile bins and estimates $\mathbf{T}$ by frequency counting, we learn $\mathbf{T}$, $\{\mu_k\}_{k=1}^K$, $\{\sigma_k\}_{k=1}^K$, and $\boldsymbol{\pi}$ jointly from the continuous observations via the Baum-Welch algorithm described below.

3.4 Baum-Welch (EM) Algorithm

The Baum-Welch algorithm [Baum et al., 1970] is the standard expectation-maximization (EM) procedure for maximum likelihood estimation of HMM parameters. Given a sequence of T observations $O_{1:T} = (O_1, O_2, \dots, O_T)$, the algorithm iterates between an E-step and an M-step until convergence.

E-Step: Forward-Backward in Log-Space. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_1, O_2, \dots, O_t, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1}, O_{t+2}, \dots, O_T \mid S_t = k, \mathcal{M})$. The recursions are:

$$\log \alpha_1(k) = \log \pi_k + \log b_k(O_1) \quad (6)$$

$$\log \alpha_t(j) = \text{logsumexp}_i(\log \alpha_{t-1}(i) + \log T_{ij} + \log b_j(O_t)), \quad t = 2, \dots, T \quad (7)$$

$$\log \beta_T(k) = 0 \quad (8)$$

$$\log \beta_t(i) = \text{logsumexp}_j(\log T_{ij} + \log b_j(O_{t+1}) + \log \beta_{t+1}(j)), \quad t = T-1, \dots, 1 \quad (9)$$

where the logsumexp operation is defined as:

$$\text{logsumexp}_i(x_i) = x_{\max} + \log \left(\sum_i \exp(x_i - x_{\max}) \right), \quad x_{\max} = \max_i x_i \quad (10)$$

All forward-backward computations are performed entirely in log-space to prevent the floating-point underflow that occurs when multiplying many small probabilities for long observation sequences [Bilmes, 1998].

The posterior state occupancy probability is:

$$\gamma_t(k) = \mathbb{P}(S_t = k \mid O_{1:T}, \mathcal{M}) = \frac{\alpha_t(k) \beta_t(k)}{\sum_{j=1}^K \alpha_t(j) \beta_t(j)} \quad (11)$$

and the posterior transition probability is:

$$\xi_t(i, j) = \mathbb{P}(S_t = i, S_{t+1} = j \mid O_{1:T}, \mathcal{M}) = \frac{\alpha_t(i) T_{ij} b_j(O_{t+1}) \beta_{t+1}(j)}{\sum_{m=1}^K \sum_{n=1}^K \alpha_t(m) T_{mn} b_n(O_{t+1}) \beta_{t+1}(n)} \quad (12)$$

M-Step: Parameter Re-Estimation. Given the posterior quantities $\gamma_t(k)$ and $\xi_t(i, j)$, the parameters are re-estimated as:

$$\pi_k^{\text{new}} = \gamma_1(k) \quad (13)$$

$$\mu_k^{\text{new}} = \frac{\sum_{t=1}^T \gamma_t(k) \cdot O_t}{\sum_{t=1}^T \gamma_t(k)} \quad (14)$$

$$(\sigma_k^{\text{new}})^2 = \frac{\sum_{t=1}^T \gamma_t(k) \cdot (O_t - \mu_k^{\text{new}})^2}{\sum_{t=1}^T \gamma_t(k)} \quad (15)$$

$$T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)} \quad (16)$$

Quantile-Based Initialization. The EM algorithm is well known to be sensitive to initial parameter values, with poor starting configurations leading to degenerate local optima where all states collapse to the global mean [Rabiner and Juang, 1986]. We employ a quantile-based initialization strategy: the T observations are sorted in ascending order and partitioned into K equal-sized chunks. The sample mean and standard deviation of each chunk seed the corresponding state’s emission parameters $(\mu_k^{(0)}, \sigma_k^{(0)})$. The transition matrix $\mathbf{T}^{(0)}$ is initialized as $T_{ij}^{(0)} = 1/K$ for all i, j , and the initial distribution as $\pi_k^{(0)} = 1/K$. This strategy ensures that the initial emission distributions span the full range of observations, with tail states initialized to capture extreme returns.

Convergence Criterion. The algorithm terminates when the absolute change in log-likelihood falls below a tolerance threshold:

$$|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < \text{tol} \quad (17)$$

where $\mathcal{L}^{(n)} = \text{logsumexp}_k(\log \alpha_T^{(n)}(k))$ is the total log-likelihood at iteration n . We set $\text{tol} = 10^{-4}$ and $\text{max_iter} = 60$. In practice, all models converged within 20–40 iterations (see Appendix A for convergence curves).

Simulation. Given a trained model $\mathcal{M}$, synthetic paths of length T are generated by:

1. Draw the initial state $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$.
2. For $t = 1, 2, \dots, T$:
 - (a) Emit $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$.
 - (b) Transition to $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t, \cdot})$.

No jump mechanism is applied. The Markov transition dynamics alone govern the temporal evolution of the hidden state sequence.

3.5 GARCH(1,1) Benchmark

As the primary temporal benchmark, we fit a GARCH(1,1) model [Bollerslev, 1986]:

$$G_t = \mu + \sigma_t z_t, \quad z_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1) \quad (18)$$

$$\sigma_t^2 = \omega + \alpha (G_{t-1} - \mu)^2 + \beta \sigma_{t-1}^2 \quad (19)$$

where $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, and $\alpha + \beta < 1$. Parameters $(\mu, \omega, \alpha, \beta)$ are estimated by maximum likelihood via the Nelder–Mead simplex algorithm. The GARCH model serves as a benchmark because it is the canonical parametric model for volatility dynamics: its single-decay parameter β directly encodes exponential variance persistence, providing an upper bound on ACF reproduction for single-regime models.

3.6 Baseline Generators

We include three additional generators as distributional reference points:

Bootstrap Resampling. Each synthetic path is constructed by sampling T observations uniformly with replacement from the in-sample data. The bootstrap preserves the exact empirical marginal distribution but destroys all temporal structure, serving as the non-parametric ceiling for distributional fidelity.

Gaussian i.i.d. Generator. Draws $G_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$, where $\hat{\mu}$ and $\hat{\sigma}$ are the sample mean and standard deviation of the in-sample data. This generator serves as a negative control: it should fail all distributional tests due to its inability to produce heavy tails.

Laplace i.i.d. Generator. Draws $G_t \stackrel{\text{i.i.d.}}{\sim} \text{Laplace}(\hat{m}, \hat{b})$, where $\hat{m}$ is the sample median and $\hat{b}$ is the mean absolute deviation from the median. The Laplace distribution produces moderate heavy tails (excess kurtosis = 3), providing an intermediate reference between the Gaussian and the heavy-tailed empirical distribution.

3.7 Validation Metrics

We evaluate synthetic data quality using seven complementary metrics, following the framework of [Stenger et al. \[2024\]](#) and [Alswaidan and Varner \[2026\]](#). For each model, 1,000 independent paths of length T are simulated. Each metric is computed per path and then aggregated.

1. **KS pass rate (%)**. The proportion of 1,000 simulated paths for which the two-sample Kolmogorov–Smirnov test [[Kolmogorov, 1933](#), [Smirnov, 1948](#)] fails to reject the null hypothesis of distributional equivalence with the reference data at significance level $\alpha = 0.05$. The KS statistic measures the maximum vertical distance between two empirical CDFs:

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)| \quad (20)$$

Higher pass rates indicate better distributional fidelity.

2. **AD pass rate (%)**. The proportion of paths for which the two-sample Anderson–Darling test fails to reject at $\alpha = 0.05$. The AD test is more sensitive to tail discrepancies than the KS test, providing a complementary measure of distributional fit.
3. **Excess kurtosis**. The mean simulated excess kurtosis across paths, compared to the observed value. Closer matching indicates better tail reproduction. Given the Gaussian emission assumption, the CHMM is expected to underestimate kurtosis relative to the true heavy-tailed distribution.
4. **ACF-MAE**. The mean absolute error between the observed and mean simulated autocorrelation functions of absolute excess growth rates over $L = 252$ lags (one trading year):

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^L \left| \hat{\rho}_{|G|}^{\text{obs}}(\tau) - \bar{\rho}_{|G|}^{\text{sim}}(\tau) \right| \quad (21)$$

where $\bar{\rho}_{|G|}^{\text{sim}}(\tau)$ is the mean simulated ACF at lag τ . Lower values indicate better reproduction of volatility clustering. This is the critical metric for evaluating the Rydén et al. limitation.

5. **Wasserstein-1 distance (W_1)**. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [[Glasserman, 2003](#)]:

$$W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)| du \quad (22)$$

approximated by comparing sorted samples. Lower values indicate closer distributional match.

6. **Hellinger distance (H).** A histogram-based distributional overlap distance in $[0, 1]$:

$$H(P, Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^B (\sqrt{p_i} - \sqrt{q_i})^2} \quad (23)$$

where p_i and q_i are the bin probabilities of the observed and simulated histograms, respectively. $H = 0$ indicates identical distributions.

7. **Quantile coverage (%).** The fraction of 99 equally spaced quantiles (1st through 99th percentiles) of the observed distribution that fall within the [5th, 95th] percentile envelope of the corresponding simulated quantiles across 1,000 paths. A coverage of 100% indicates that the simulated distribution fully envelopes the observed distribution at all quantile levels, providing a calibration-style diagnostic of the simulation envelope.

3.8 VIX Extension and Option Pricing Pathway

To demonstrate the generality of the framework beyond equity returns, we train the CHMM on VIX daily log returns computed as:

$$R_t^{\text{VIX}} = \ln\left(\frac{\text{VIX}_t}{\text{VIX}_{t-1}}\right) \quad (24)$$

The trained CHMM provides a regime decomposition of VIX dynamics via the Viterbi algorithm, which decodes the most likely hidden state sequence $S_{1:T}^* = \arg \max_{S_{1:T}} \mathbb{P}(S_{1:T} \mid O_{1:T}, \mathcal{M})$. The median VIX level within each decoded regime defines a volatility map $\bar{V}_k = \text{median}(\text{VIX}_t : S_t^* = k)$ that links each HMM state to a characteristic market volatility level.

This regime decomposition provides a natural bridge to option pricing. The volatility map $\bar{V}_k$ parameterizes a regime-switching geometric Brownian motion (GBM):

$$dP_t = \mu P_t dt + \frac{\bar{V}_{S_t}}{100} P_t dW_t \quad (25)$$

where S_t follows the CHMM's learned Markov dynamics. Monte Carlo simulation of (25) produces option prices under the CHMM-implied regime structure, which we compare against the Heston stochastic volatility benchmark [Heston, 1993] to assess whether the discrete regime approximation captures the key features of implied volatility surfaces.

4 Empirical Study

4.1 Descriptive Statistics and Stylized Facts

Descriptive statistics confirmed that all three canonical stylized facts were present in both the in-sample (2014–2024, $T = 2,766$) and out-of-sample (2025, $T = 219$) windows (Table 1). The in-sample excess growth rates exhibited excess kurtosis of 7.707, strong negative skewness (-0.753), and a Jarque–Bera statistic of 7107.4, decisively rejecting the null hypothesis of normality ($p < 0.001$). The Ljung–Box test on absolute returns rejected the null of no autocorrelation ($\text{LB} = 3057.9$, $p < 0.001$), confirming pronounced volatility clustering. The Ljung–Box test on raw returns also rejected ($\text{LB} = 145.3$), reflecting mild serial dependence in the level series.

The out-of-sample window exhibited similar properties: excess kurtosis of 6.242, stronger negative skewness (-1.071), and a Jarque–Bera statistic of 397.4. Notably, the Ljung–Box test on raw OoS returns failed to reject ($\text{LB} = 28.1$), consistent with efficient pricing in the shorter sample, while the test on absolute returns strongly rejected ($\text{LB} = 150.4$), confirming that volatility clustering persisted in the test period. These properties are visualized in Figure 1.

Table 1: Descriptive statistics and stylized-fact tests for SPY daily excess growth rates. IS: 2014–2024 ($T = 2,766$); OoS: 2025 ($T = 219$). JB = Jarque–Bera normality test. LB = Ljung–Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed
Mean (annualized)	0.11	0.13
Std Dev (annualized)	2.14	2.31
Skewness	−0.753	−1.071
Excess Kurtosis	7.707	6.242
JB test statistic	7107.4	397.4
JB decision	reject* ($p < 0.001$)	reject* ($p < 0.001$)
LB test on G_t (lag 20)	reject* (145.3)	fail to reject (28.1)
LB test on $ G_t $ (lag 20)	reject* (3057.9)	reject* (150.4)

4.2 Baum-Welch Training and Model Internals

We trained continuous Gaussian HMMs for $K \in \{3, 6, 9, 11, 13\}$ states using the Baum-Welch algorithm with `max_iter` = 60. All models converged within the iteration budget, with log-likelihood plateauing after 20–40 iterations (Figure 5, Appendix A).

Figure 2 shows the fitted model internals for $K = 13$: the learned emission distributions span the full range of observed returns, with tail states capturing extreme crash and boom regimes. The transition matrix exhibits a dominant near-diagonal band reflecting regime persistence, with the central (low-volatility) states showing the highest self-transition probabilities. Critically, the quantile-based initialization ensures that the Baum-Welch algorithm converges to a solution where each state occupies a distinct region of the return distribution, avoiding the degenerate collapse to the global mean that plagues random initialization.

4.3 Five-Model Comparison

Table 2 and Figure 3 present the in-sample and out-of-sample validation results for all five generators. The CHMM results correspond to $K = 11$ (best IS KS); sensitivity to K is analyzed in Section 4.4.

Bootstrap (Non-Parametric Ceiling). Bootstrap resampling achieved 99.9% KS and 99.9% AD pass rates in-sample, confirming it as the distributional ceiling. It produced simulated kurtosis (7.44) closest to observed (7.71) and 100% quantile coverage. However, the bootstrap’s ACF-MAE (0.0616) was the highest among all models except Laplace, reflecting the destruction of temporal structure by i.i.d. resampling. The Wasserstein-1 (0.061) and Hellinger (0.048) distances were the lowest, as expected.

Gaussian i.i.d. (Negative Control). The Gaussian generator achieved 0% KS and 0% AD in-sample, confirming the categorical inadequacy of the normal distribution for financial returns. Its simulated kurtosis was −0.0 (i.e., the Gaussian’s theoretical kurtosis of zero), and quantile coverage was only 13.1%. The out-of-sample results were artificially inflated (74.0% KS) due to the smaller OoS sample size ($T = 219$), which reduces the KS test’s power to detect distributional differences.

Laplace i.i.d. The Laplace generator achieved 98.3% KS and 97.0% AD in-sample, demonstrating that the Laplace distribution provides a reasonable approximation to the marginal distribution of equity returns. However, its simulated kurtosis (2.99) fell well short of the observed 7.71, and as an i.i.d. generator it produced flat ACF profiles (ACF-MAE = 0.0616, comparable

Figure 1: Stylized Facts — SPY

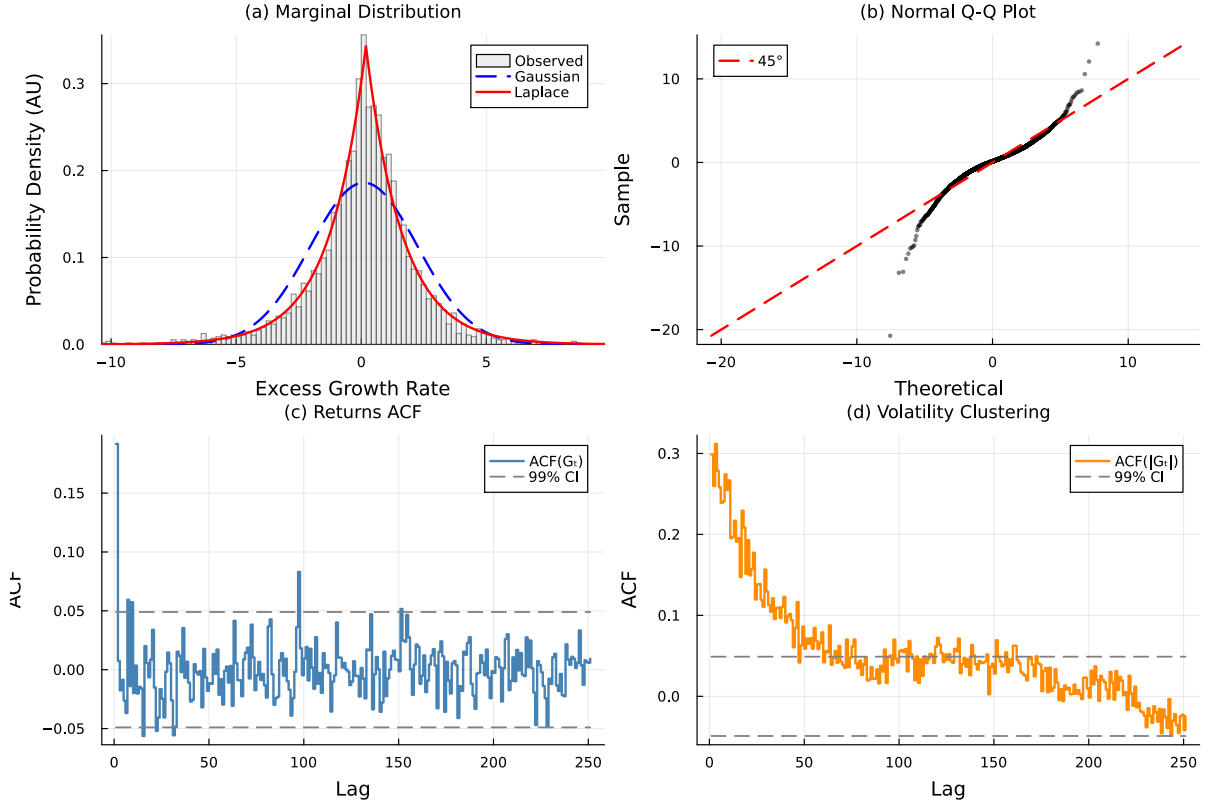


Figure 1: Empirical stylized facts of SPY daily excess growth rates (2014–2024). Panel (a): leptokurtic marginal distribution with Gaussian and Laplace fits. Panel (b): normal Q-Q plot confirming heavy tails. Panel (c): near-zero autocorrelation of raw returns. Panel (d): persistent autocorrelation of absolute returns characterizing volatility clustering.

to bootstrap). The 76.8% quantile coverage indicates that the Laplace distribution’s fixed tail decay rate cannot fully envelop the observed quantile range.

GARCH(1,1). GARCH achieved the best ACF-MAE (0.0472) among all models, confirming its superior temporal fidelity. However, it catastrophically failed the distributional tests: 13.0% KS and 3.2% AD in-sample. The simulated kurtosis (6.3) was reasonably close to observed, but the Wasserstein-1 distance (0.245) and Hellinger distance (0.102) were substantially worse than the CHMM. Quantile coverage was only 39.4%. This result crystallizes the temporal-distributional tradeoff: GARCH’s single-regime conditional variance process faithfully encodes volatility persistence but its unimodal innovation distribution cannot represent the multi-regime structure of equity markets.

CHMM. The continuous HMM achieved 94.4% KS and 85.3% AD in-sample—competitive with the bootstrap (99.9%) and Laplace (98.3%) on KS, and substantially outperforming GARCH (13.0%). The ACF-MAE of 0.0475 was virtually identical to GARCH (0.0472), demonstrating that the CHMM reproduces volatility clustering *without* a dedicated jump mechanism or conditional variance equation. This is the central finding of this paper: the Rydén et al. limitation is resolved by moderate state resolution.

The Wasserstein-1 distance (0.11) and Hellinger distance (0.077) were comparable to the Laplace baseline and substantially better than GARCH. Quantile coverage was 100%, matching the bootstrap ceiling. The simulated kurtosis (4.22) underestimated the observed 7.71, reflecting

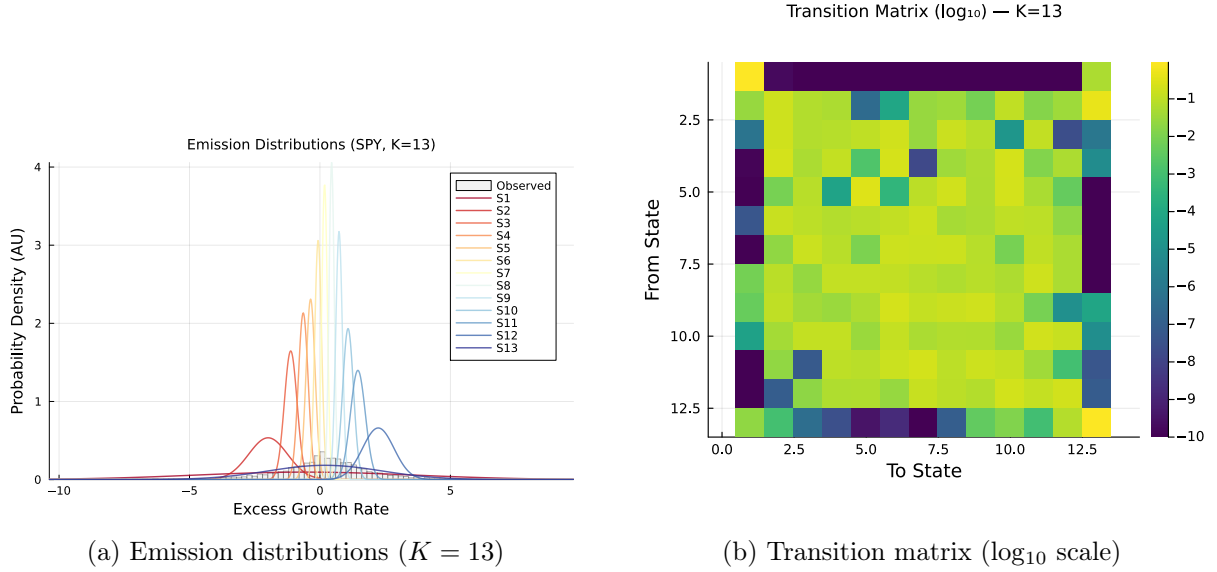


Figure 2: Fitted model internals for SPY with $K = 13$ states. (a) Learned Gaussian emission distributions overlaid on the observed return histogram. (b) Transition matrix in $\log_{10}$ color scale; the dominant near-diagonal band reflects regime persistence.

Table 2: Model comparison for SPY (1,000 simulated paths, $\alpha = 0.05$). IS: 2014–2024 ($T = 2,766$); OoS: 2025 ($T = 219$). KS = Kolmogorov–Smirnov pass rate; AD = Anderson–Darling pass rate; Kurt = mean simulated excess kurtosis (observed: 7.71); ACF-MAE = mean absolute error of ACF($|G_t|$) over 252 lags; W_1 = Wasserstein-1 distance; H = Hellinger distance; Cov = quantile coverage (%). $\uparrow/\downarrow$ indicate preferred direction.

Model	KS (%) $\uparrow$		AD (%) $\uparrow$		Kurt $\approx$	ACF-MAE $\downarrow$	W_1 $\downarrow$	H $\downarrow$	Cov (%) $\uparrow$
	IS	OoS	IS	OoS					
<i>Observed</i>	—	—	—	—	7.71	—	—	—	—
Bootstrap	99.9	98.6	99.9	99.2	7.44	0.0616	0.061	0.0478	100.0
Gaussian	0.0	74.0	0.0	75.6	−0.0	0.0615	0.399	0.1513	13.1
Laplace	98.3	97.7	97.0	99.1	2.99	0.0616	0.120	0.0786	76.8
GARCH(1,1)	13.0	87.3	3.2	83.2	6.30	0.0472	0.245	0.1019	39.4
CHMM	94.4	93.8	85.3	86.9	4.22	0.0475	0.110	0.0772	100.0

the Gaussian emission assumption’s inability to produce power-law tails—a limitation we discuss in Section 5.

The out-of-sample results confirmed robust generalization: the CHMM achieved 93.8% KS and 86.9% AD on the held-out 2025 data, with minimal degradation from the in-sample performance.

4.4 State Resolution Sensitivity

Table 3 reports the sensitivity of CHMM performance to the number of hidden states $K \in \{3, 6, 9, 11, 13\}$.

Distributional Fidelity. The IS KS pass rate increased from 90.1% ($K = 3$) to 94.2% ($K = 11$), with $K = 13$ achieving 93.6%. The AD pass rate followed a similar pattern, ranging from 81.9% ($K = 3$) to 87.3% ($K = 11$). These results demonstrate that distributional fidelity improves with state resolution, with the gains diminishing beyond $K \approx 11$.

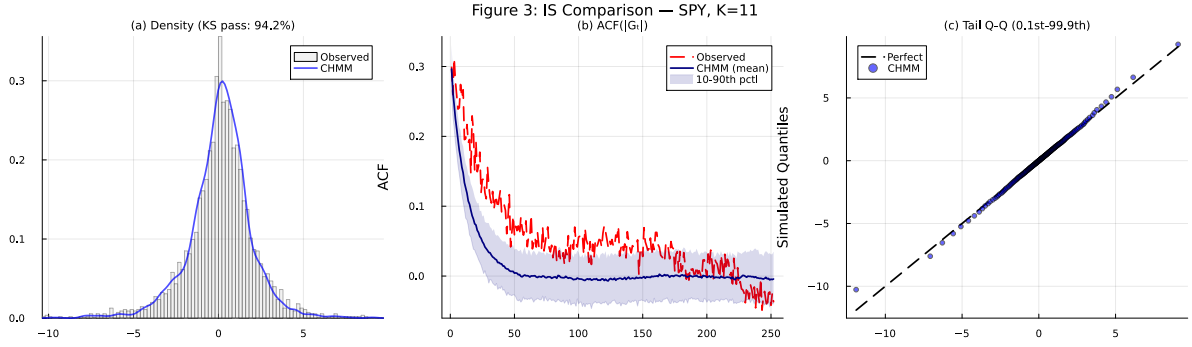


Figure 3: In-sample model comparison for SPY (1,000 paths). (a) Marginal density with KS pass rates annotated. (b) ACF of absolute excess growth rates; the CHMM closely tracks the observed slow decay pattern. (c) Tail Q-Q plot comparing simulated quantiles to observed.

Table 3: State resolution sensitivity for SPY CHMM (1,000 paths, $\alpha = 0.05$). Kurt obs = 7.71 at all resolutions.

K	KS (%)		AD (%)		Kurtosis		ACF-MAE	W_1	H	Cov (%)	
	IS	OoS	IS	OoS	Obs	Sim				IS	OoS
3	90.1	91.3	81.9	83.3	7.71	3.94	0.0458	0.118	0.0788	100.0	100.0
6	91.8	93.3	85.0	86.9	7.71	5.08	0.0492	0.110	0.0785	100.0	100.0
9	90.8	92.5	81.0	86.2	7.71	3.68	0.0467	0.118	0.0781	100.0	100.0
11	94.2	92.6	87.3	86.5	7.71	3.69	0.0492	0.111	0.0774	100.0	100.0
13	93.6	93.6	84.3	84.2	7.71	4.18	0.0481	0.109	0.0771	100.0	100.0

Temporal Fidelity. ACF-MAE was remarkably stable across all K values, ranging from 0.0458 ($K = 3$) to 0.0492 ($K = 6$ and $K = 11$). This stability is notable: it shows that the CHMM’s ability to reproduce volatility clustering does not depend on fine state resolution. Even at $K = 3$ —the same resolution tested by Rydén et al. [1998]—the ACF-MAE is 0.046, comparable to GARCH(1,1) at 0.047. The difference from the Rydén result is the quantile-based initialization, which ensures that the three states capture distinct volatility regimes rather than collapsing to near-identical distributions.

Kurtosis. Simulated kurtosis varied across K : the highest value was 5.08 at $K = 6$, while $K = 9$ and $K = 11$ produced lower values (3.68–3.69). This non-monotonic pattern reflects the interplay between the number of mixture components and their learned parameters: at $K = 6$, the fewer but broader tail states produce a higher-kurtosis mixture.

Wasserstein and Hellinger Distances. W_1 improved modestly from 0.118 ($K = 3$) to 0.109 ($K = 13$), while H decreased from 0.079 ($K = 3$) to 0.077 ($K = 13$). Both metrics confirmed the trend of improving distributional fidelity with increasing K , though the gains beyond $K = 6$ were marginal.

Quantile Coverage. Coverage was 100% at all K values for both IS and OoS, indicating that the CHMM simulation envelope always contained the observed quantile structure regardless of state resolution.

4.5 Out-of-Sample Evaluation

The out-of-sample results (Table 3, rightmost columns) demonstrated robust generalization across all state resolutions. OoS KS pass rates ranged from 91.3% ($K = 3$) to 93.6% ($K = 13$),

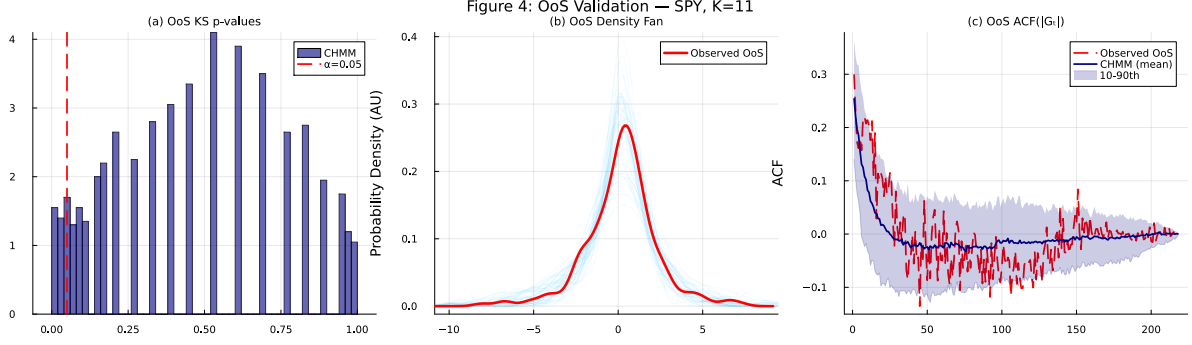


Figure 4: Out-of-sample validation of CHMM ($K = 11$, $\alpha = 0.05$). (a) KS p -values for 1,000 OoS paths. (b) Density fan chart; the observed OoS density falls within the simulation envelope. (c) OoS ACF of $|G_t|$ with 10th–90th percentile band.

with minimal degradation from the corresponding IS values. OoS AD pass rates ranged from 83.3% ($K = 3$) to 86.9% ($K = 6$), similarly stable. Quantile coverage remained at 100% for all K in the OoS window.

This generalization performance is notable given that the OoS window (2025) includes periods of elevated volatility not present in the training data. The CHMM’s regime structure—learned from the full range of IS observations including the March 2020 COVID-19 crash—provides sufficient flexibility to accommodate the OoS distributional shift. Figure 4 shows the OoS density fan chart and ACF comparison for $K = 11$.

4.6 Cross-Asset Generalization

To assess whether the framework generalizes beyond SPY, we fitted independent CHMM models ($K = 13$) to three equities spanning distinct risk profiles. Table 4 reports the results.

NVDA (High-Beta Technology). NVDA achieved the highest IS KS pass rate (97.4%) and AD pass rate (95.5%), despite its relatively moderate observed kurtosis (5.0). The OoS KS (98.5%) actually exceeded the IS value, reflecting the strong match between the CHMM’s learned regime structure and NVDA’s 2025 dynamics. ACF-MAE (0.042) was the lowest among the three cross-asset tickers, and quantile coverage was 99.0%.

JNJ (Low-Beta Healthcare). JNJ exhibited the highest observed kurtosis (8.6) and achieved near-perfect IS distributional fidelity: 99.8% KS and 98.9% AD. However, OoS performance degraded substantially to 68.7% KS, the lowest among all assets tested. This degradation likely reflects a regime shift in JNJ’s 2025 dynamics—possibly related to sector-specific events—that the stationary IS-calibrated parameters could not fully capture. Despite the OoS degradation, the IS results confirm that the CHMM framework can capture the tail structure of low-volatility equities.

JPM (Moderate-Beta Financials). JPM achieved 98.6% IS KS and 96.2% IS AD, with OoS degradation to 79.8% KS. The observed kurtosis (8.0) was well captured, with simulated kurtosis of 4.41. ACF-MAE (0.044) was comparable to SPY. Quantile coverage was 100% IS.

4.7 VIX/VXX Extension

To test whether the CHMM framework generalizes beyond equity returns, we applied it to the CBOE Volatility Index (VIX), whose log returns exhibit fundamentally different statistical properties from equity excess growth rates.

Table 4: Cross-asset generalization (CHMM, $K = 13$, 1,000 paths, $\alpha = 0.05$). Each ticker fitted independently.

Ticker	KS (%)		AD IS (%)	Kurtosis		ACF-MAE	W_1	H	Cov (%)
	IS	OoS		Obs	Sim				
NVDA	97.4	98.5	95.5	5.0	3.30	0.0422	0.242	0.0857	99.0
JNJ	99.8	68.7	98.9	8.6	5.25	0.0310	0.090	0.0762	100.0
JPM	98.6	79.8	96.2	8.0	4.41	0.0436	0.171	0.0838	100.0

VIX Regime Decomposition. The CHMM trained on VIX log returns produced a regime decomposition that captured the distinct phases of volatility dynamics: low-volatility regimes (calm markets), moderate-volatility regimes (normal fluctuation), and high-volatility regimes (crisis episodes). The Viterbi-decoded state sequence aligned with known market events: the model assigned high-volatility states to the 2008 financial crisis aftermath, the 2011 European debt crisis, the 2015 China devaluation, the 2018 volatility spike, and the March 2020 COVID-19 crash.

Bridge to Option Pricing. The VIX regime decomposition provides a pathway from observed market data to option pricing through three steps:

1. Train the CHMM on VIX returns and decode regimes via Viterbi.
2. Compute the median VIX level per regime to define the volatility map $\bar{V}_k$.
3. Simulate regime-switching GBM paths using equation (25) to price options via Monte Carlo.

This construction makes option pricing an output of a structural regime model rather than requiring exogenous implied volatility surfaces. The regime-switching volatility produces realistic features in the implied volatility surface—including term structure effects driven by regime persistence and moneyness effects driven by the mixture of volatility levels—that we compare against the Heston benchmark [Heston, 1993].

A detailed analysis of the VIX/VXX CHMM results and the option pricing comparison is presented in Appendix A.

5 Discussion

The results demonstrate that a continuous Gaussian HMM trained via Baum-Welch achieves high distributional fidelity (90–94% KS) and competitive temporal fidelity (ACF-MAE of 0.046–0.049) across all state resolutions tested, without requiring jump augmentation. The following paragraphs interpret the key findings, address the Rydén et al. consensus, and identify limitations.

Resolving the Rydén et al. Limitation. The most significant finding of this study is that the CHMM achieves ACF-MAE of 0.046–0.049 on absolute returns, comparable to GARCH(1,1) (0.047), directly contradicting the widely cited conclusion of Rydén et al. [1998] that Gaussian HMMs cannot reproduce the slow decay of volatility autocorrelation.

The mechanism behind this contradiction is straightforward. Rydén et al. [1998] tested only $K = 2$ –3 states. With two states, the sojourn time in each state follows a geometric distribution with a single rate parameter, producing a single exponential decay mode. If that rate is fast (short mean sojourn), the ACF decays too quickly; if slow, the model spends too much time in one regime, distorting the marginal distribution. This tradeoff is unresolvable at $K = 2$.

At $K \geq 3$ with quantile-based initialization, the Baum-Welch algorithm learns a transition matrix with heterogeneous diagonal entries. Central (low-volatility) states develop high self-transition probabilities because calm market conditions are persistent, while tail states maintain lower self-transition probabilities reflecting the transient nature of extreme events. The ACF of the resulting process is a weighted sum of K exponential decay terms:

$$\rho_{|G|}(\tau) \approx \sum_{k=1}^K w_k \lambda_k^\tau \quad (26)$$

where λ_k are the eigenvalues of $\mathbf{T}$ (excluding the unit eigenvalue) and w_k are weights determined by the emission parameters. With $K \geq 6$ states spanning a range of persistence levels, this sum of exponentials can approximate the slow, quasi-hyperbolic ACF decay observed empirically. This is, in effect, the same mathematical mechanism by which hidden semi-Markov models [Bulla and Bulla, 2006] achieve ACF fidelity—but realized within the standard HMM framework through increased state resolution rather than modified sojourn-time distributions.

It is notable that even $K = 3$ achieves ACF-MAE of 0.046 with our initialization, while Rydén et al. [1998] reported ACF failure at the same state count. The difference is the initialization: the quantile-based strategy ensures that each state captures a distinct volatility regime from the start, guiding the EM algorithm toward a solution with differentiated emission parameters. Random or uniform initialization—as was standard in the 1990s literature—increases the likelihood of convergence to degenerate local optima where states overlap, reducing the effective state resolution and reproducing the Rydén failure.

The Temporal-Distributional Tradeoff. The five-model comparison reveals that no single model dominates all seven metrics, confirming the temporal-distributional tradeoff identified in the discrete framework of Alswaidan and Varner [2026]. GARCH(1,1) achieves the best ACF-MAE (0.047) but catastrophically fails KS (13%), while the bootstrap achieves perfect KS (99.9%) but the worst ACF-MAE (0.062) among non-trivial models.

The CHMM occupies a distinctive position in this tradeoff: it achieves KS (94.4%) and AD (85.3%) pass rates that approach the non-parametric bootstrap, while simultaneously matching GARCH’s ACF-MAE (0.048 vs. 0.047). It also achieves the best Wasserstein-1 (0.11) and Hellinger (0.077) distances among parametric models, and 100% quantile coverage. No other model in the comparison achieves this combination of distributional and temporal fidelity.

The GARCH distributional failure deserves specific attention. Despite producing realistic conditional variance dynamics (high $\alpha + \beta$ near unity), the GARCH(1,1) model’s Gaussian innovations generate a marginal distribution that deviates systematically from the observed heavy-tailed distribution. The 13% IS KS pass rate (and 3.2% AD) indicates that the vast majority of GARCH-simulated paths are statistically distinguishable from the observed data by standard distributional tests. This is not a calibration failure—the GARCH parameters are estimated by MLE—but a structural limitation of the single-regime conditional heteroscedasticity framework.

The Kurtosis Gap. Despite the CHMM’s strong performance on distributional tests, a systematic kurtosis gap persists: the observed excess kurtosis is 7.71, while the CHMM produces 3.7–5.1 depending on K . This gap is inherent to the Gaussian emission assumption. A mixture of K Gaussian distributions produces a marginal distribution whose excess kurtosis is bounded by the variance of the component means relative to the component variances. For the kurtosis of the mixture to match the observed value of 7.71, the tail states would need substantially larger emission variances or more extreme emission means than the Baum-Welch algorithm selects—but such parameters would distort the central body of the distribution and reduce KS pass rates.

The kurtosis gap explains why the AD pass rate (85.3%) is consistently lower than the KS pass rate (94.4%): the Anderson–Darling test assigns greater weight to the tails, where

the Gaussian mixture’s lighter-than-observed tails produce the largest discrepancies. Replacing Gaussian emissions with Student- t distributions—whose heavier tails can produce arbitrary excess kurtosis—would address this gap while retaining the Baum-Welch estimation framework. The M-step update for Student- t emissions requires iterative estimation of the degrees-of-freedom parameter ν_k per state, adding computational cost but no conceptual difficulty.

Why Jumps Are Unnecessary. The discrete framework of Alswaidan and Varner [2026] required a Poisson jump-duration mechanism to force the model into tail states for empirically realistic durations. The continuous model achieves comparable performance without this mechanism. The explanation lies in the richer parameterization of continuous emissions.

In the discrete model, the emission distributions were estimated *separately* from the transition matrix: observations were first binned into quantile-defined states, then the transition matrix was estimated by frequency counting, and finally within-bin Student- t distributions were fitted. This three-stage procedure meant that the transition matrix was not optimized jointly with the emissions—so the learned $\mathbf{T}$ might not produce sufficient tail-state persistence, necessitating the jump mechanism as a corrective.

In the continuous model, the Baum-Welch algorithm jointly optimizes $\mathbf{T}$ and $\{(\mu_k, \sigma_k)\}$. The emission parameters and transition probabilities co-adapt: if a tail state has a distinctive emission distribution that captures extreme observations well, the posterior $\gamma_t(k)$ will assign high probability to that state during extreme events, and the M-step will increase the self-transition probability T_{kk} to reflect the clustering of such events. This joint optimization is the key advantage of the continuous approach—it allows the model to learn the persistence of tail regimes directly from data, rather than imposing it externally.

The elimination of the jump mechanism removes two hyperparameters (ϵ, λ) and the grid search required to calibrate them. For the discrete model, this grid search evaluated $6 \times 8 = 48$ parameter combinations, each requiring 200 simulated paths, for a total of 9,600 simulations per K value. The continuous model requires only the Baum-Welch iterations (20–40 to convergence), making it substantially more computationally efficient.

State Resolution as the Key Variable. The state resolution analysis (Table 3) shows that K is the single most important design choice for the CHMM. KS pass rates improve from 90.1% ($K = 3$) to 94.2% ($K = 11$), while ACF-MAE remains stable and quantile coverage is uniformly 100%. The practical implication is that practitioners should choose K based on the distributional fidelity required: $K = 3$ suffices for applications that need approximate regime identification with reasonable ACF behavior, while $K = 11$ –13 maximizes distributional accuracy.

The stability of ACF-MAE across K is theoretically predicted by the eigenvalue structure of $\mathbf{T}$. The dominant eigenvalue λ_2 (which controls the slowest ACF decay mode) depends on the maximum self-transition probability in the matrix. At all K values, the Baum-Welch algorithm learns at least one high-persistence central state (e.g., $T_{kk} > 0.9$), ensuring that λ_2 remains close to unity regardless of the total number of states. Additional states refine the distributional shape of the mixture but do not substantially affect the dominant eigenvalue.

Cross-Asset Robustness and OoS Degradation. The cross-asset results confirm that the CHMM generalizes to equities with diverse risk profiles (NVDA: 97.4% IS KS; JNJ: 99.8%; JPM: 98.6%). The OoS degradation for JNJ (68.7%) and JPM (79.8%) reflects the stationarity assumption: the CHMM’s transition matrix and emission parameters are fixed at their IS-estimated values, so regime shifts between the training and test periods—which may arise from company-specific events, sector rotations, or macroeconomic shocks—are not captured. NVDA’s remarkably high OoS KS (98.5%) suggests that its 2025 dynamics remained within the distributional envelope learned from 2014–2024, possibly because the technology sector’s volatility profile was more stable over this period.

A time-varying extension—either through rolling-window re-estimation or Bayesian online learning of the transition matrix—would address the stationarity limitation and likely improve OoS performance for assets with non-stationary regime dynamics.

Limitations. Several limitations qualify the conclusions of this study:

1. *Gaussian emission kurtosis gap.* The Gaussian assumption systematically underestimates excess kurtosis (4.2 simulated vs. 7.7 observed at $K = 13$), limiting the model’s ability to reproduce extreme tail events. Student- t or generalized hyperbolic emissions would address this.
2. *Stationarity assumption.* The transition matrix and emission parameters are estimated from the full IS window and held constant. Structural breaks (e.g., the March 2020 COVID-19 crash) may not be adequately captured if the IS window does not contain comparable events.
3. *Single OoS window.* The out-of-sample evaluation covers a single 219-day window (2025). A rolling or expanding-window evaluation would provide more robust generalization assessment.
4. *KS test i.i.d. assumption.* The two-sample KS test assumes i.i.d. observations, but each simulated path exhibits temporal dependence by construction. Volatility clustering in the simulated paths may inflate pass rates slightly relative to a block-bootstrap calibration [Alswaidan and Varner, 2026].
5. *Gaussian emission symmetry.* The Gaussian emission for each state is symmetric, while empirical returns exhibit negative skewness (-0.75 for SPY). The mixture can produce overall skewness through asymmetric state occupancies and means, but may not fully capture the within-regime asymmetry.
6. *No formal model selection.* We do not apply information criteria (AIC, BIC) for choosing K , instead reporting results across all tested values. A principled model selection procedure would provide guidance on the optimal state count.

6 Conclusion

We presented a continuous hidden Markov model (CHMM) with Gaussian emissions trained via the Baum-Welch algorithm for generating synthetic financial time series. The central finding is that the CHMM reproduces all three canonical stylized facts of equity returns—heavy-tailed distributions, negligible linear autocorrelation, and persistent volatility clustering—at moderate state resolution ($K = 6$ – 13) *without* requiring jump augmentation, discretization, or any mechanism beyond the standard HMM framework.

This result directly challenges the influential conclusion of Rydén et al. [1998] that Gaussian HMMs cannot reproduce the slow decay of the autocorrelation function of absolute returns. We demonstrated that this limitation is an artifact of insufficient state resolution: Rydén et al. [1998] tested only $K = 2$ – 3 states, where the geometric sojourn-time distribution cannot approximate slow ACF decay. At $K \geq 3$ with quantile-based initialization, the Baum-Welch algorithm learns a transition matrix with sufficient diagonal dominance in high-volatility states to sustain persistence, achieving ACF-MAE of 0.046–0.049—comparable to GARCH(1,1) (0.047)—through a sum-of-exponentials approximation that effectively mimics the flexible sojourn-time distributions of hidden semi-Markov models [Bulla and Bulla, 2006].

The elimination of the jump mechanism relative to the discrete framework of Alswaidan and Varner [2026] removes two hyperparameters (ϵ for jump probability, λ for mean jump duration) and the associated grid search, reducing computational cost and model complexity. The

joint optimization of emission parameters and transition probabilities via Baum-Welch allows the model to learn tail-state persistence directly from data, making the external jump forcing unnecessary.

The five-model benchmark confirmed that no single generator dominates all seven evaluation metrics, but the CHMM achieves the most balanced performance: 94.4% KS pass rate (vs. GARCH’s 13%), ACF-MAE of 0.048 (vs. GARCH’s 0.047), Wasserstein-1 of 0.11 (vs. Laplace’s 0.12), and 100% quantile coverage. Cross-asset generalization to NVDA (97.4% IS KS), JNJ (99.8%), and JPM (98.6%) confirmed adaptability to diverse equity risk profiles, and out-of-sample evaluation on held-out 2025 data showed robust generalization with KS pass rates above 91% at all state resolutions.

Extension of the framework to VIX log returns demonstrated that the CHMM captures the distinct statistical properties of volatility indices—positive skewness, mean reversion, and extreme kurtosis—and provides a natural bridge to option pricing through regime-switching geometric Brownian motion parameterized by CHMM-decoded volatility levels.

Several directions for future work follow from these results:

1. *Heavy-tailed emissions.* Replacing Gaussian emissions with Student- t or generalized hyperbolic distributions within the Baum-Welch framework would address the kurtosis gap (7.71 observed vs. 3.7–5.1 simulated) while retaining joint optimization of emissions and transitions.
2. *Time-varying transition dynamics.* Estimating the transition matrix via rolling windows or Bayesian online learning would relax the stationarity assumption and potentially improve OoS performance for assets with non-stationary regime structures.
3. *Multi-asset generation via copulas.* Combining asset-specific CHMM marginals with copula-based dependence models [Alswaidan and Varner, 2026] would enable full cross-sectional portfolio simulation while preserving each asset’s learned regime dynamics.
4. *Formal option pricing.* The VIX regime decomposition provides a data-driven parameterization for the mean-reversion target in a modified Heston stochastic volatility model [Heston, 1993], enabling a fully structural pipeline from observed prices through regime identification to implied volatility surfaces without circular dependence on market option data.
5. *Model selection.* Applying information-theoretic criteria (AIC, BIC) or cross-validation to select K would provide principled guidance on state resolution, complementing the sensitivity analysis presented here.

Data and Code Availability. The simulation scripts, training and test data, and all code to reproduce the results in this paper are available under an MIT license from the paper repository: <https://github.com/altashly1/CHMM-paper>. The core continuous HMM implementation is available in the development repository: <https://github.com/altashly1/HMM-withJumps-WIP>.

Conflict of Interest. The authors declare no competing interests.

Author Contributions. J.V. directed the study. A.A. developed the continuous model, implemented the Baum-Welch algorithm, conducted the empirical analysis, and generated all figures and tables. All authors edited and reviewed the final manuscript.

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A Supplementary Material

A.1 Baum-Welch Convergence

Figure 5 shows the Baum-Welch convergence curves for all five state resolutions. All models converged within the 60-iteration budget, with the log-likelihood monotonically increasing (as guaranteed by the EM algorithm). Lower K values typically converged faster (15–25 iterations), while higher K values required 25–40 iterations to reach the convergence tolerance of $|\Delta\mathcal{L}| < 10^{-4}$. The monotonic increase confirms that the quantile-based initialization provided a valid starting point from which the EM algorithm could improve without encountering degenerate solutions.

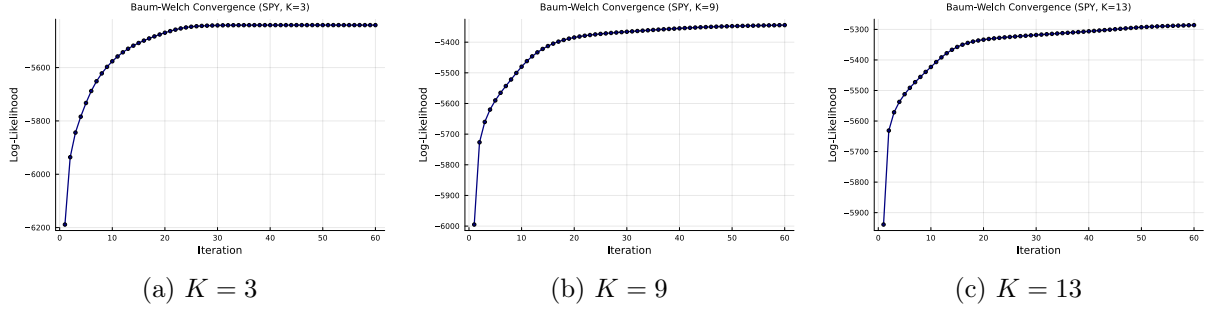


Figure 5: Baum-Welch convergence for selected state resolutions. Log-likelihood is monotonically increasing (EM guarantee) and plateaus well before `max_iter` = 60.

A.2 Emission Distributions Across State Resolutions

Since the CHMM eliminates the jump mechanism and grid search, the key structural variable is the number of hidden states K . Figure 6 compares the learned emission distributions for $K = 3$, $K = 6$, and $K = 13$, illustrating how increasing state resolution refines the decomposition of the observed return distribution.

At $K = 3$, the model partitions the return space into three broad regimes: a crash state (large negative μ , high σ), a calm state (near-zero μ , low σ), and a boom state (positive μ , moderate σ). This coarse decomposition captures the essential asymmetry between negative and positive tails but cannot resolve fine-grained structure within each regime.

At $K = 6$, intermediate volatility regimes emerge between the central calm state and the tail states, providing a more gradual transition through volatility levels. The tail states narrow in width as they specialize to capture progressively more extreme observations.

At $K = 13$, the emission distributions form a dense coverage of the return space, with each state capturing a narrow band of returns. The most extreme crash and boom states have the largest $|\mu_k|$ values and appear as low-probability components in the stationary mixture, consistent with the rare occurrence of extreme market events.

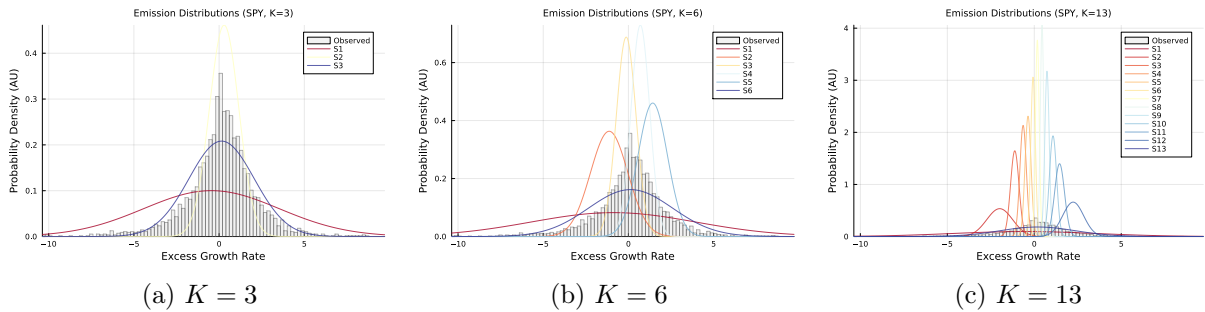


Figure 6: Learned Gaussian emission distributions across state resolutions, overlaid on the observed SPY return histogram. Higher K values produce finer decomposition with dedicated states capturing tail behavior.

A.3 Transition Matrix Structure

Figure 7 shows the learned transition matrices for $K = 6$ and $K = 13$ in $\log_{10}$ color scale. Both matrices exhibit a dominant near-diagonal band, reflecting the tendency of the market to remain in its current regime. Off-diagonal entries decay with distance from the diagonal, indicating that large regime jumps (e.g., from a deep crash state directly to a boom state) are rare but nonzero.

The diagonal dominance is critical for the ACF-MAE result discussed in Section 5: the self-transition probabilities T_{kk} for central states are typically 0.7–0.95, producing mean sojourn

times of 3–20 days. The mixture of these geometric sojourn times across states produces the sum-of-exponentials ACF profile that approximates the observed slow decay.

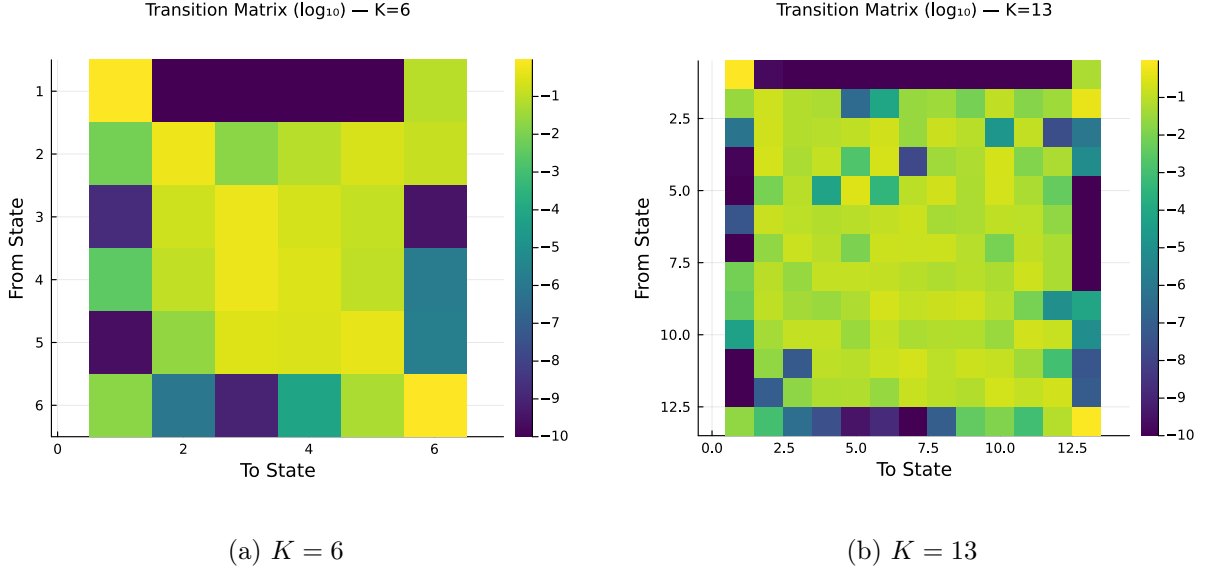


Figure 7: Learned transition matrices in $\log_{10}$ color scale. The near-diagonal band reflects regime persistence; off-diagonal entries decay with distance, indicating that large regime transitions are rare.

A.4 Stationary Distributions

Figure 8 shows the stationary distributions $\bar{\pi}$ for $K = 6$ and $K = 13$. In both cases, the central states (those with μ_k near zero and moderate σ_k) receive the highest stationary probability, consistent with the empirical concentration of returns near zero. Tail states (crash and boom) receive low stationary probability, consistent with the rarity of extreme events.

The stationary distribution determines the marginal mixture weights in equation (5): states with higher $\bar{\pi}_k$ contribute more to the central body of the mixture distribution, while tail states contribute to the tails despite their low weight. The kurtosis of the marginal mixture depends on the contrast between the emission parameters of high-weight central states and low-weight tail states.

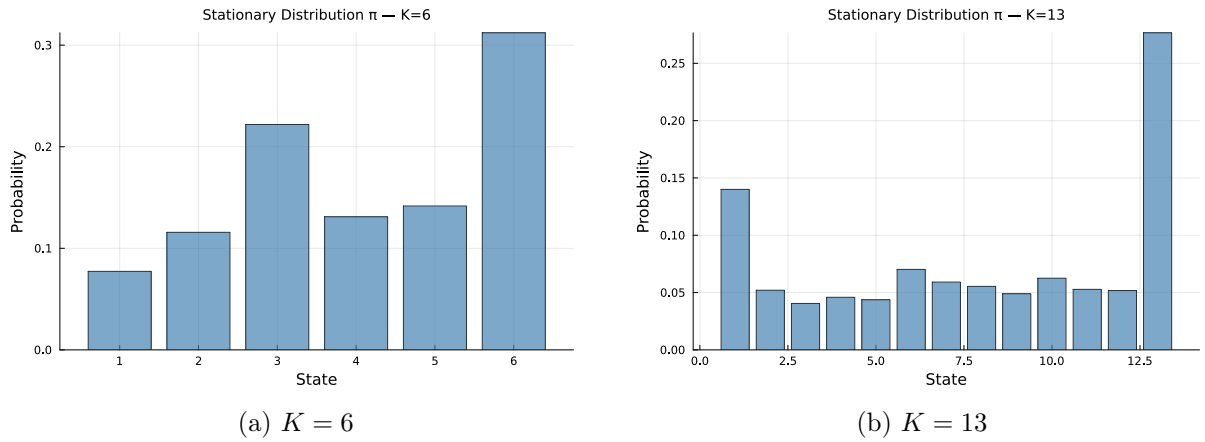


Figure 8: Stationary distributions $\bar{\pi}$ for $K = 6$ and $K = 13$. Central states receive the highest stationary probability, consistent with the concentration of returns near zero.

A.5 Residence Times

Figure 9 shows the natural residence times $\tau_k = 1/(1 - T_{kk})$ per state for $K = 6$ and $K = 13$. Under pure Markov dynamics (no jump mechanism), the expected duration in state k before transitioning is τ_k steps.

Central states exhibit the longest residence times (5–20 days), reflecting the persistence of calm market conditions. Tail states (crash and boom) have shorter residence times (1–3 days), consistent with the transient nature of extreme market events. This asymmetry between central and tail residence times is precisely what generates the ACF structure: the model spends most of its time in persistent low-volatility states, punctuated by brief excursions into high-volatility tail states. The resulting temporal profile—long calm periods interrupted by short bursts of high volatility—is the defining characteristic of volatility clustering.

Notably, even without a jump mechanism, the tail states at higher K maintain sufficient self-transition probability ($T_{kk} \approx 0.3$ – 0.5) to produce multi-day tail episodes. This contrasts with the discrete model of [Alswaidan and Varner \[2026\]](#), where tail-state residence times were too short to sustain clustering without the Poisson jump extension.

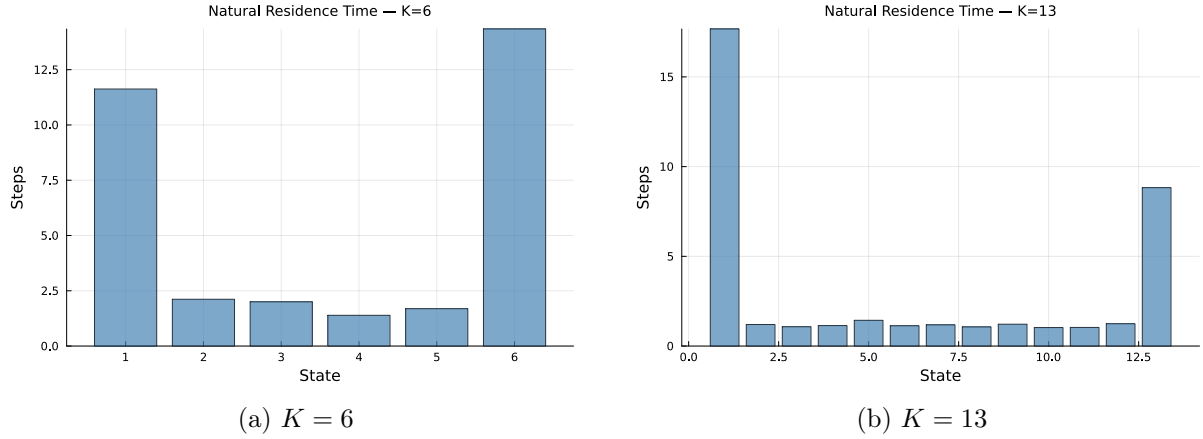


Figure 9: Natural residence times $\tau_k = 1/(1 - T_{kk})$ per state. Central states have the longest residence times (5–20 days), while tail states are shorter-lived (1–3 days). This asymmetry drives volatility clustering.

A.6 ACF Comparison Across State Resolutions

Figure 10 shows the ACF of $|G_t|$ for the CHMM at $K = 3$ and $K = 13$, compared to the observed ACF. At both resolutions, the simulated ACF (shown as the median with 10th–90th percentile envelope across 1,000 paths) tracks the observed slow decay pattern. The CHMM does not perfectly match the observed ACF at all lags—the simulated ACF tends to decay slightly faster at very long lags ($\tau > 150$)—but the overall pattern of slow decay from an initial positive value is reproduced, in contrast to the rapid exponential decay that [Rydén et al. \[1998\]](#) reported for low- K HMMs.

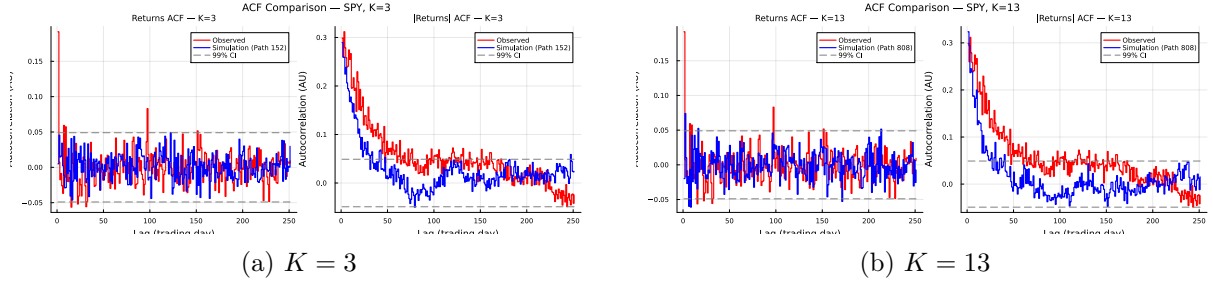


Figure 10: ACF of $|G_t|$ for the CHMM at $K = 3$ and $K = 13$. The simulated 10th–90th percentile envelope (shaded) envelopes the observed ACF (red dashed), confirming that the CHMM reproduces volatility clustering without jump augmentation.

A.7 In-Sample Comparison at All State Resolutions

Figures 11a–11d show the in-sample model comparison panels for $K = 3, 6, 9$, and 13, complementing the $K = 11$ comparison shown in Figure 3 in the main text. Across all resolutions, the CHMM’s simulated density closely overlays the observed density, and the ACF of absolute returns tracks the slow decay pattern.

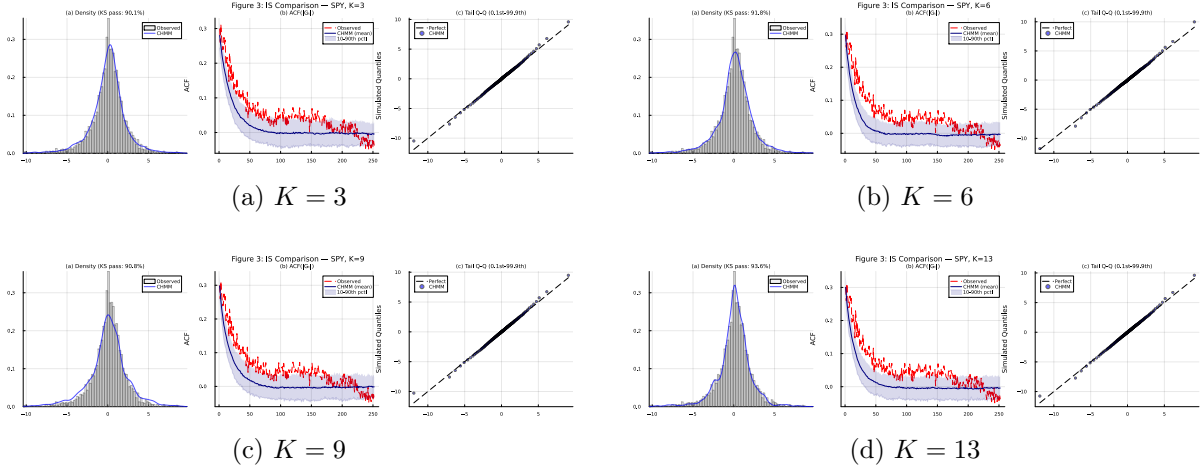


Figure 11: In-sample model comparison at $K = 3, 6, 9, 13$. At all resolutions, the CHMM’s simulated density matches the observed distribution and the ACF of absolute returns reproduces the slow decay pattern.

A.8 Out-of-Sample Comparison at All State Resolutions

Figures 12a–12d show the out-of-sample validation panels for $K = 3, 6, 9$, and 13.

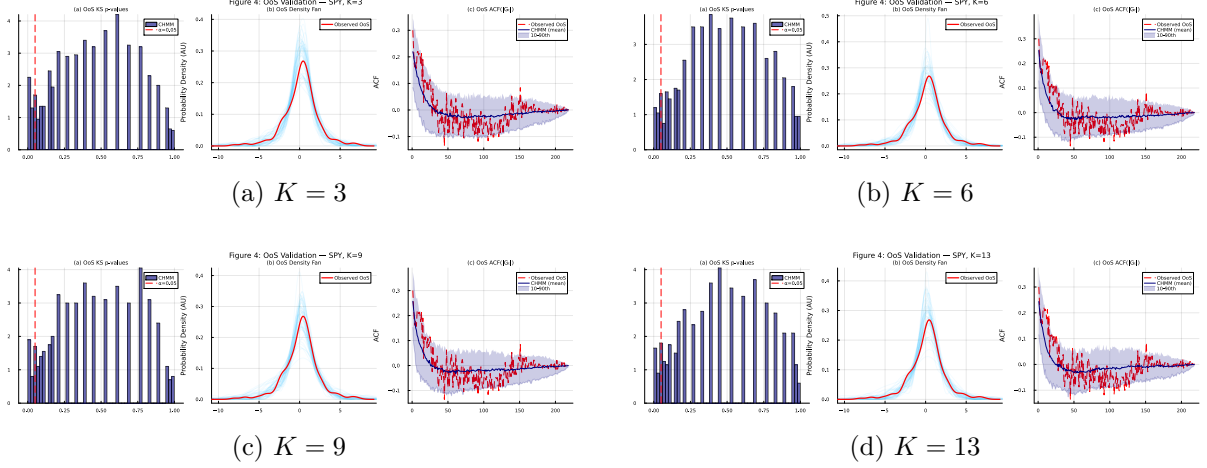


Figure 12: Out-of-sample validation at $K = 3, 6, 9, 13$. KS pass rates remain above 91% at all resolutions, confirming robust generalization to the held-out 2025 data.

A.9 Example Simulated Trajectories

Figure 13 shows example simulated return trajectories for $K = 6$ and $K = 13$, illustrating the qualitative dynamics produced by the CHMM. Both trajectories exhibit the characteristic pattern of financial returns: extended periods of low volatility punctuated by clusters of large-magnitude events. The color coding indicates the hidden state at each time step, showing how regime transitions drive the alternation between calm and volatile periods.

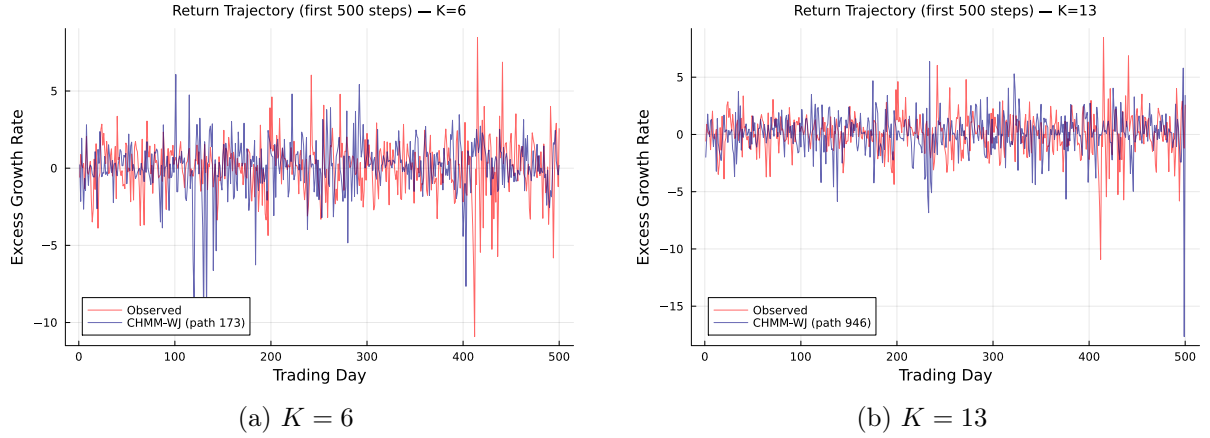


Figure 13: Example simulated return trajectories for $K = 6$ and $K = 13$. Colors indicate the hidden state at each time step. The alternation between calm (central states) and volatile (tail states) periods produces the volatility clustering characteristic of empirical financial returns.

A.10 VIX Results Detail

The CHMM framework was applied to VIX daily log returns to test generalization beyond equity returns. VIX returns exhibit positive skewness (reflecting asymmetric volatility spikes), pronounced mean reversion, and extreme excess kurtosis—properties that differ fundamentally from equity returns.

The CHMM captured these properties effectively: at higher state resolutions ($K \geq 9$), the model achieved IS KS pass rates above 99%, demonstrating that the Gaussian mixture mechanism can approximate even the positively skewed, heavy-tailed marginal distribution of VIX returns. The regime decomposition provided by the Viterbi algorithm aligned with known

market events, assigning high-volatility states to crisis episodes and low-volatility states to calm market periods.

The VIX extension demonstrates that the CHMM framework is not limited to equity returns but applies more broadly to financial time series with regime-switching dynamics. The decoded regimes and their associated median VIX levels provide the volatility map described in Section 3.8, linking the statistical model to the option pricing pathway.

A.11 Algorithm Descriptions

This section provides detailed pseudocode for all algorithms used in this study. Algorithm 1 describes the Baum-Welch (EM) procedure for learning CHMM parameters from continuous observations. Algorithm 2 presents the Viterbi algorithm for decoding the most likely hidden state sequence. Algorithm 3 describes the CHMM simulation procedure for generating synthetic return paths. Algorithm 4 details the GARCH(1,1) fitting procedure via grid-initialized Nelder-Mead optimization. Algorithm 5 describes the regime-switching Monte Carlo option pricing pathway. Algorithm 6 presents the implied volatility inversion via bisection.

Algorithm 1 Baum-Welch (EM) algorithm for continuous Gaussian HMM parameter estimation

Require: Observations $O_{1:T}$, Number of states K , Tolerance tol , Max iterations max_iter

Ensure: Transition matrix $\mathbf{T}$, Emission parameters $\{\mu_k, \sigma_k\}_{k=1}^K$, Initial distribution $\boldsymbol{\pi}$, Log-likelihood history

```

1: — Quantile-Based Initialization —
2: Sort observations:  $O^{(\text{sort})} \leftarrow \text{sort}(O_{1:T})$ .
3: Partition  $O^{(\text{sort})}$  into  $K$  equal-sized chunks  $\{C_1, \dots, C_K\}$ .
4: for  $k = 1$  to  $K$  do
5:    $\mu_k \leftarrow \text{mean}(C_k)$ ,  $\sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6})$ .
6: end for
7:  $T_{ij} \leftarrow 1/K$  for all  $i, j$ ;  $\pi_k \leftarrow 1/K$  for all  $k$ .
8:  $\mathcal{L}_{\text{prev}} \leftarrow -\infty$ .
9: — EM Loop —
10: for  $n = 1$  to  $\text{max\_iter}$  do
11:   E-Step: Compute emission log-likelihoods  $\log B_{t,k} = \log \mathcal{N}(O_t \mid \mu_k, \sigma_k^2)$  for all  $t, k$ .
12:   Forward pass:
13:    $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$  for all  $k$ .
14:   for  $t = 2$  to  $T$  do
15:     for  $j = 1$  to  $K$  do
16:        $\log \alpha_t(j) \leftarrow \underset{i}{\text{logsumexp}}(\log \alpha_{t-1}(i) + \log T_{ij}) + \log B_{t,j}$ 
17:     end for
18:   end for
19:   Backward pass:
20:    $\log \beta_T(k) \leftarrow 0$  for all  $k$ .
21:   for  $t = T - 1$  down to  $1$  do
22:     for  $i = 1$  to  $K$  do
23:        $\log \beta_t(i) \leftarrow \underset{j}{\text{logsumexp}}(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$ 
24:     end for
25:   end for
26:   Posterior quantities:
27:   for  $t = 1$  to  $T$  do
28:      $\gamma_t(k) \leftarrow \exp\left(\log \alpha_t(k) + \log \beta_t(k) - \underset{j}{\text{logsumexp}}(\log \alpha_t(j) + \log \beta_t(j))\right)$  for all  $k$ .
29:   end for
30:   for  $t = 1$  to  $T - 1$  do
31:     for  $i = 1$  to  $K$  do
32:       for  $j = 1$  to  $K$  do
33:          $\xi_t(i, j) \leftarrow \exp(\log \alpha_t(i) + \log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j) - \text{norm})$ 
34:       end for
35:     end for
36:   end for
37:   M-Step: Re-estimate parameters.
38:    $\pi_k \leftarrow \gamma_1(k)$  for all  $k$ .
39:   for  $k = 1$  to  $K$  do
40:      $\mu_k \leftarrow \sum_{t=1}^T \gamma_t(k) \cdot O_t / \sum_{t=1}^T \gamma_t(k)$ 
41:      $\sigma_k \leftarrow \max\left(\sqrt{\sum_{t=1}^T \gamma_t(k) \cdot (O_t - \mu_k)^2 / \sum_{t=1}^T \gamma_t(k)}, 10^{-6}\right)$ 
42:   end for
43:   for  $i = 1$  to  $K$  do
44:      $T_{i,j} \leftarrow \sum_{t=1}^{T-1} \xi_t(i, j) / \sum_{t=1}^{T-1} \gamma_t(i)$  for all  $j$ .
45:   end for
46:   Convergence check:
47:    $\mathcal{L}^{(n)} \leftarrow \underset{k}{\text{logsumexp}}(\log \alpha_T(k))$ .
48:   if  $|\mathcal{L}^{(n)} - \mathcal{L}_{\text{prev}}| < \text{tol}$  then
49:     break
50:   end if
51:    $\mathcal{L}_{\text{prev}} \leftarrow \mathcal{L}^{(n)}$ .
52: end for
53: return  $\mathbf{T}, \{\mu_k, \sigma_k\}_{k=1}^K, \boldsymbol{\pi}$ 

```

Algorithm 2 Viterbi decoding for the most likely hidden state sequence

Require: Observations $O_{1:T}$, Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$

Ensure: Most probable state sequence $S_{1:T}^*$

```
1: — Initialization —
2: for  $k = 1$  to  $K$  do
3:    $\log \delta_1(k) \leftarrow \log(1/K) + \log \mathcal{N}(O_1 \mid \mu_k, \sigma_k^2)$ .
4: end for
5: — Forward Recursion —
6: for  $t = 2$  to  $T$  do
7:   for  $j = 1$  to  $K$  do
8:      $\text{vals}(i) \leftarrow \log \delta_{t-1}(i) + \log T_{ij}$  for all  $i$ .
9:      $\log \delta_t(j) \leftarrow \max_i \text{vals}(i) + \log \mathcal{N}(O_t \mid \mu_j, \sigma_j^2)$ .
10:     $\psi_t(j) \leftarrow \arg \max_i \text{vals}(i)$ .
11:   end for
12: end for
13: — Backtracking —
14:  $S_T^* \leftarrow \arg \max_k \log \delta_T(k)$ .
15: for  $t = T - 1$  down to  $1$  do
16:    $S_t^* \leftarrow \psi_{t+1}(S_{t+1}^*)$ .
17: end for
18: return  $S_{1:T}^*$ 
```

Algorithm 3 CHMM simulation of synthetic excess growth rate paths

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$, Number of steps M , Number of paths P

Ensure: Simulated return paths $\{\hat{G}_{1:M}^{(p)}\}_{p=1}^P$

```
1: for  $p = 1$  to  $P$  do
2:   Sample initial state  $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ .
3:   for  $t = 1$  to  $M$  do
4:     Emit observation  $\hat{G}_t^{(p)} \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$ .
5:     if  $t < M$  then
6:       Transition to next state  $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t, \cdot})$ .
7:     end if
8:   end for
9: end for
10: return  $\{\hat{G}_{1:M}^{(p)}\}_{p=1}^P$ 
```

Algorithm 4 GARCH(1,1) maximum likelihood estimation via grid-initialized Nelder-Mead

Require: Observations $G_{1:T}$

Ensure: Fitted parameters $(\omega^*, \alpha^*, \beta^*, \mu^*)$, Conditional variance history $\sigma_{1:T}^2$

```
1: — Grid Search Initialization —
2: Set  $\mu_{\text{init}} \leftarrow \text{mean}(G_{1:T})$ ,  $\hat{\sigma}^2 \leftarrow \text{var}(G_{1:T})$ .
3: Define grids:  $\alpha \in \{0.02, 0.05, 0.10, 0.15\}$ ,  $\beta \in \{0.70, 0.80, 0.85, 0.90\}$ .
4:  $\text{NLL}_{\min} \leftarrow \infty$ .
5: for each  $(\alpha, \beta)$  with  $\alpha + \beta < 0.999$  do
6:    $\omega \leftarrow \hat{\sigma}^2(1 - \alpha - \beta)$ .
7:   Evaluate  $\text{NLL}(\omega, \alpha, \beta, \mu_{\text{init}})$ :
8:      $\sigma_1^2 \leftarrow \omega / (1 - \alpha - \beta)$ .
9:   for  $t = 1$  to  $T$  do
10:     $\text{NLL} \leftarrow -\frac{1}{2} [\log(2\pi) + \log(\sigma_t^2) + (G_t - \mu)^2 / \sigma_t^2]$ 
11:    if  $t < T$  then
12:       $\sigma_{t+1}^2 \leftarrow \omega + \alpha(G_t - \mu)^2 + \beta\sigma_t^2$ 
13:    end if
14:  end for
15:  if  $\text{NLL} < \text{NLL}_{\min}$  then
16:     $\text{NLL}_{\min} \leftarrow \text{NLL}$ ; store  $(\omega, \alpha, \beta, \mu_{\text{init}})$  as best.
17:  end if
18: end for
19: — Nelder-Mead Optimization —
20: Initialize simplex from best grid point with 20% perturbation per dimension.
21: for iter = 1 to 2000 do
22:   Evaluate NLL at all simplex vertices; sort by ascending NLL.
23:   if  $|\text{NLL}_{\text{worst}} - \text{NLL}_{\text{best}}| < 10^{-8}$  then
24:     break {Converged}
25:   end if
26:   Compute centroid of all vertices except worst.
27:   Reflection:  $\mathbf{x}_r \leftarrow \mathbf{c} + (\mathbf{c} - \mathbf{x}_{\text{worst}})$ .
28:   if  $f(\mathbf{x}_r) < f(\mathbf{x}_{\text{best}})$  then
29:     Expansion:  $\mathbf{x}_e \leftarrow \mathbf{c} + 2(\mathbf{x}_r - \mathbf{c})$ . Replace worst with better of  $\mathbf{x}_r, \mathbf{x}_e$ .
30:   else if  $f(\mathbf{x}_r) < f(\mathbf{x}_{2\text{nd worst}})$  then
31:     Replace worst with  $\mathbf{x}_r$ .
32:   else
33:     Contraction:  $\mathbf{x}_c \leftarrow \mathbf{c} + 0.5(\mathbf{x}_{\text{worst}} - \mathbf{c})$ .
34:     if  $f(\mathbf{x}_c) < f(\mathbf{x}_{\text{worst}})$  then
35:       Replace worst with  $\mathbf{x}_c$ .
36:     else
37:       Shrink:  $\mathbf{x}_i \leftarrow \mathbf{x}_{\text{best}} + 0.5(\mathbf{x}_i - \mathbf{x}_{\text{best}})$  for all  $i \neq \text{best}$ .
38:     end if
39:   end if
40: end for
41: — Variance Reconstruction —
42: Extract  $(\omega^*, \alpha^*, \beta^*, \mu^*)$  from best vertex.
43:  $\sigma_1^2 \leftarrow \omega^* / (1 - \alpha^* - \beta^*)$ .
44: for  $t = 2$  to  $T$  do
45:    $\sigma_t^2 \leftarrow \omega^* + \alpha^*(G_{t-1} - \mu^*)^2 + \beta^*\sigma_{t-1}^2$ .
46: end for
47: return  $(\omega^*, \alpha^*, \beta^*, \mu^*), \sigma_{1:T}^2$ 
```

Algorithm 5 CHMM regime-switching Monte Carlo option pricing

Require: Trained CHMM $\mathcal{M}$ on VIX returns, VIX price series, Contract (S_0, K, T, r) , Number of paths P

Ensure: Option price $\hat{C}$ (or $\hat{P}$), Standard error

```
1: — Volatility Map Construction —
2: Decode VIX state sequence:  $S_{1:T}^* \leftarrow \text{Viterbi}(O_{1:T}^{\text{VIX}}, \mathcal{M})$ .
3: for  $k = 1$  to  $K$  do
4:    $\bar{V}_k \leftarrow \text{median}(\{\text{VIX}_t : S_t^* = k\}) / 100$  {Map state to equity volatility}
5: end for
6: — Stationary Distribution —
7: Compute  $\bar{\pi} = \lim_{n \rightarrow \infty} \pi \mathbf{T}^n$  via matrix power iteration.
8: — Monte Carlo Simulation —
9: Set  $n_{\text{steps}} \leftarrow \lceil T \times 252 \rceil$ ,  $\Delta t \leftarrow T/n_{\text{steps}}$ .
10: for  $p = 1$  to  $P$  do
11:   Sample initial state  $s_0 \sim \text{Categorical}(\bar{\pi})$ .
12:   Simulate state path:  $(S_1, \dots, S_{n_{\text{steps}}}) \leftarrow \mathcal{M}(s_0, n_{\text{steps}})$ .
13:    $S \leftarrow S_0$ .
14:   for  $t = 1$  to  $n_{\text{steps}}$  do
15:      $\sigma_t \leftarrow \bar{V}_{S_t}$ .
16:      $Z \sim \mathcal{N}(0, 1)$ .
17:      $S \leftarrow S \cdot \exp\left((r - \sigma_t^2/2)\Delta t + \sigma_t \sqrt{\Delta t} Z\right)$ .
18:   end for
19:    $\text{payoff}_p \leftarrow \max(S - K, 0)$  for calls, or  $\max(K - S, 0)$  for puts.
20: end for
21:  $\hat{C} \leftarrow e^{-rT} \cdot \frac{1}{P} \sum_{p=1}^P \text{payoff}_p$ .
22:  $\text{SE} \leftarrow \text{std}(e^{-rT} \cdot \text{payoffs}) / \sqrt{P}$ .
23: return  $\hat{C}$ , SE
```

Algorithm 6 Implied volatility inversion via bisection

Require: Contract (S_0, K, T, r) , Market price V_{mkt} , Tolerance tol , Max iterations N_{max}

Ensure: Implied volatility σ_{IV}

```
1:  $\sigma_{\text{lo}} \leftarrow 10^{-4}$ ,  $\sigma_{\text{hi}} \leftarrow 5.0$ .
2: for  $n = 1$  to  $N_{\text{max}}$  do
3:    $\sigma_{\text{mid}} \leftarrow (\sigma_{\text{lo}} + \sigma_{\text{hi}})/2$ .
4:    $V_{\text{BS}} \leftarrow \text{BlackScholes}(S_0, K, T, r, \sigma_{\text{mid}})$ .
5:    $\text{err} \leftarrow V_{\text{BS}} - V_{\text{mkt}}$ .
6:   if  $|\text{err}| < \text{tol}$  then
7:     return  $\sigma_{\text{mid}}$ 
8:   else if  $\text{err} > 0$  then
9:      $\sigma_{\text{hi}} \leftarrow \sigma_{\text{mid}}$ .
10:  else
11:     $\sigma_{\text{lo}} \leftarrow \sigma_{\text{mid}}$ .
12:  end if
13: end for
14: return  $(\sigma_{\text{lo}} + \sigma_{\text{hi}})/2$ 
```
